Engineering Economics (CSE-481) week-6 : Economic Equivalence

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Economic Equivalence: Why Time Changes Money

1 A simple question to start

Would you rather receive $1,000 today or $1,200 one year from now?

In engineering economics, we do not compare money alone. We compare **money with time**.



2 A startup decision

Build your own AI infrastructure:
$50,000 today

vs.

Use cloud services: annual payments over time



The mistake begins when we compare totals without comparing timing.

> **A dollar in 2026 is not the same economic object as a dollar in 2031.**

Key takeaway: **Economic equivalence** = translating cash flows to a common point in time.

Economic Equivalence

- Economic equivalence is the language that lets money speak across time.
- Engineers don’t compare numbers; we compare *moments* in time translated into value
- Every investment is a story told in two tenses: now and later.
- Engineers don’t just design systems — we design how money flows through time

Economic Equivalence

Imagine I offer you two deals.

Deal A: I'll give you $1,000 today.
Deal B: I'll give you $1,200 one year from now.

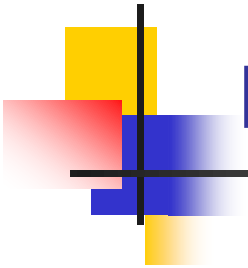
Which one would you take?"

- Most people would pause — because the answer isn't obvious. It depends on something invisible, yet incredibly powerful: **the time value of money.**

Economic Equivalence

- In engineering economics, we call this idea **Economic Equivalence** — the principle that two different cash flows can **actually be *worth the same*** if you look at them from the right point in time.

- Economic equivalence is like a *translator* between time and money. It helps us compare “now” versus “later”



Economic Equivalence

Think of it like this:
You're a software engineer at a startup.

You can either:

- Invest $50,000 now to develop your own AI infrastructure,
  or
- Pay $10,000 per year for the next six years to use a cloud provider.

Which option is cheaper?

Economic Equivalence

At first glance, you might say, “Well, $10,000 per year sounds less.”

But once we apply **economic equivalence**, we realize that a dollar next year is *not the same* as a dollar today — because money today can earn interest, be invested, or be lost to inflation.

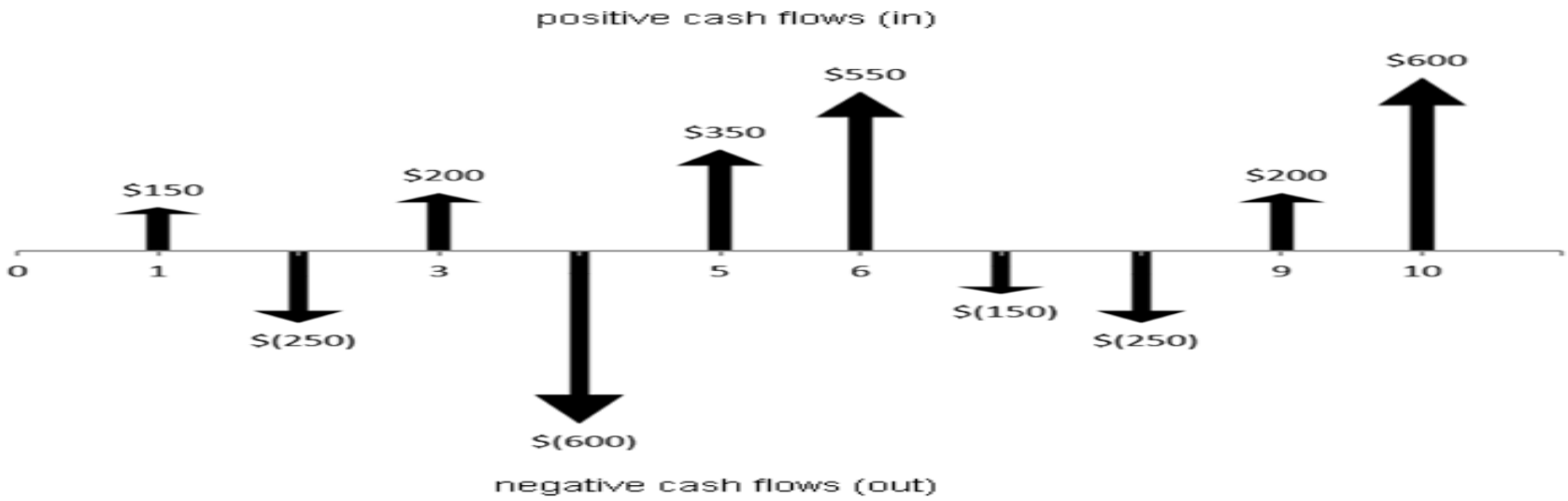
So we translate those future payments into **present value**, or convert the current investment into a **future equivalent** — and only then can we make a fair comparison.

Economic Equivalence

The technique of ***Economic Equivalence*** can answer such questions

- In engineering economics, **two things** are said to **be equivalent** if they **have the same effect**
- Thus, **two or more cash flows are equivalent** if their **time value of money are equal** at a **common point in time**
- ***Economic equivalence*** is the concept that links **different cash flows** associated with **different alternatives** so that **comparisons** based on **economic worth** can be made

Economic Equivalence



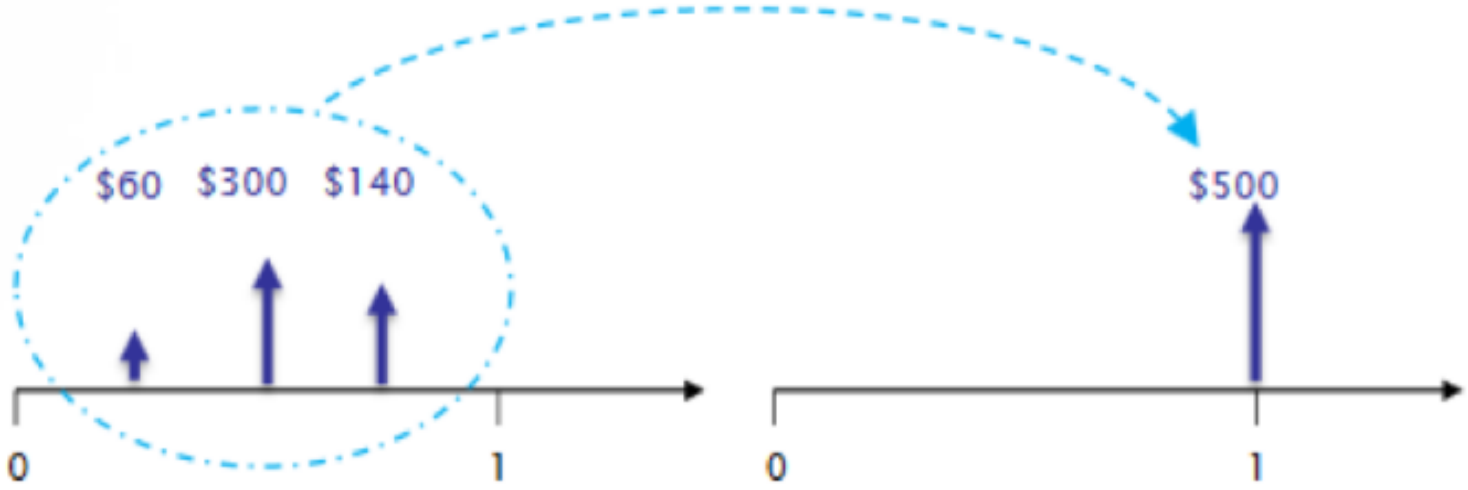
Cash Flow Diagrams are simple graphical representations of financial transactions. The diagrams consist of arrows, such as in the diagram shown above.

- **Upward arrows** represent **positive** cash flows, also known as **inflows**, **income**, or **receipts**.
- **Downward arrows** represent **negative** cash flows, also known as **outflows**, **disbursements**, or **expenses**.
- **Each arrow** represents the **net** cash flow *in that period* (receipts – disbursements). There is only one **cash flow arrow** for each period representing this net value.

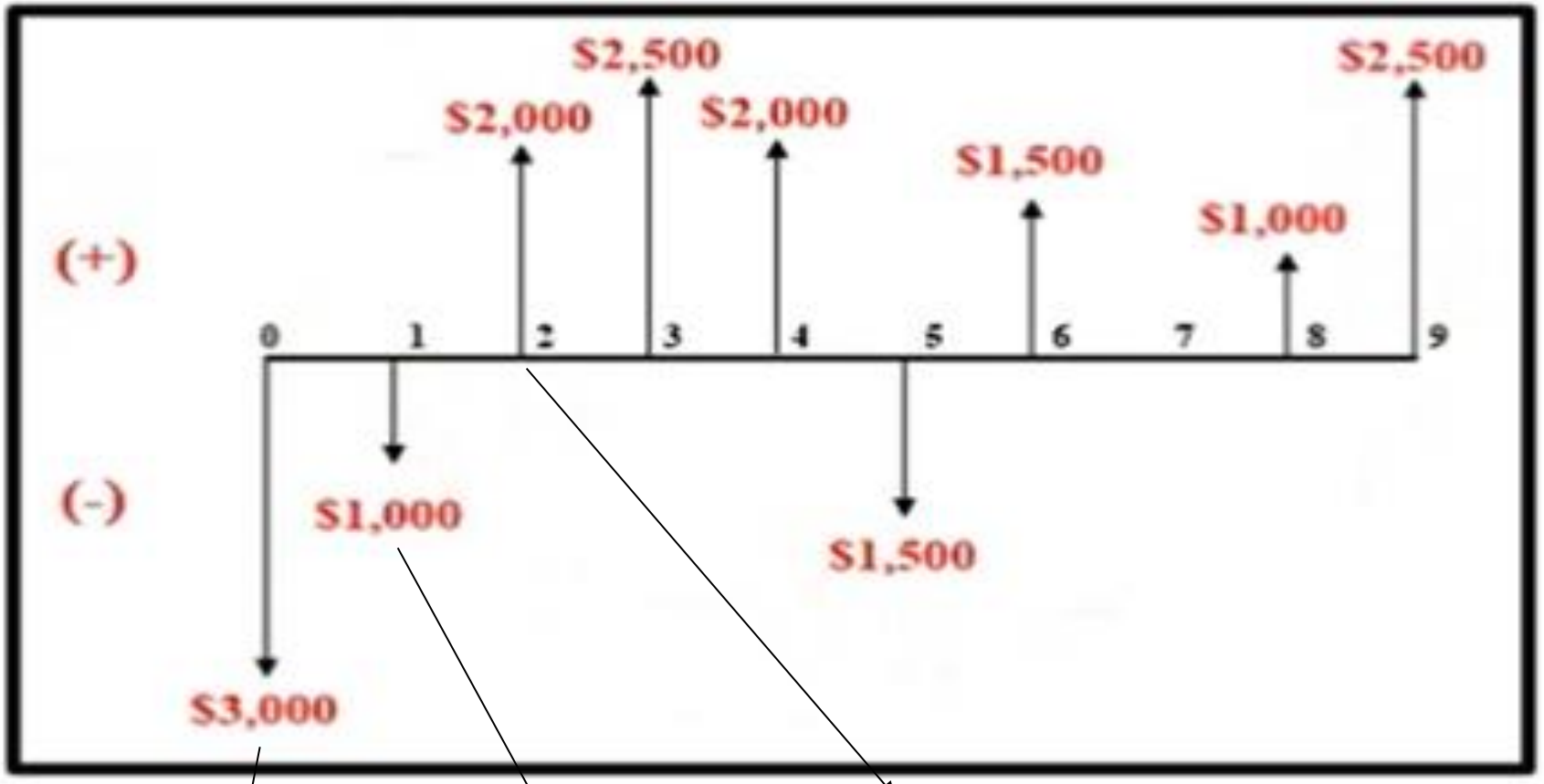
Economic Equivalence

End-of-Period Convention

In practice, cash flows can occur at **any time within a period**. However, for simplicity, we commonly assume the **end-of-period convention** – the **assumption that all cash flows occurring within a period are moved to the end of the period.** The following figure graphically demonstrates the end-of-period convention.



Economic Equivalence



an expense or initial cost of $3.000

expense of $1000 after 1 year

revenue of $2000 at the end of the 2th year

Equivalent to @ i = 15%

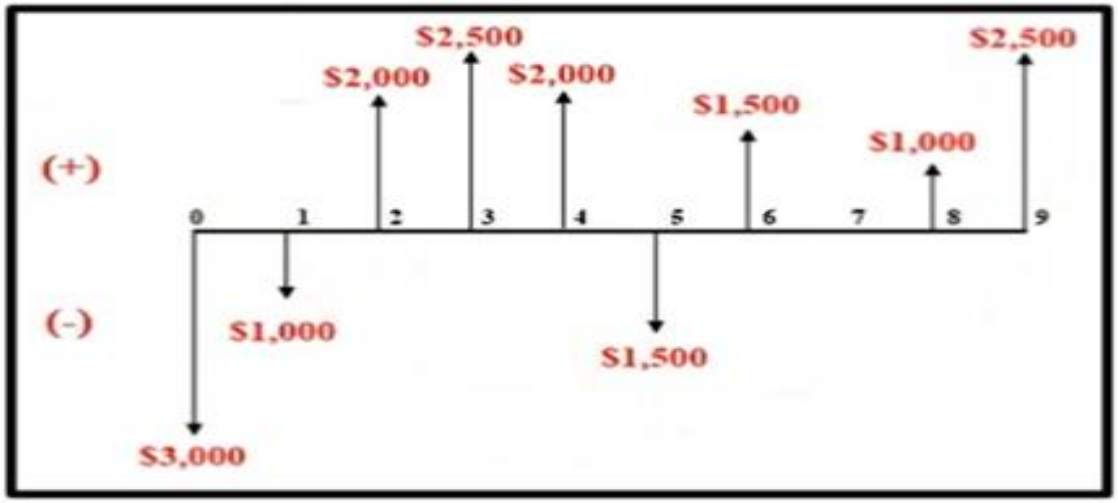
Economic Equivalence



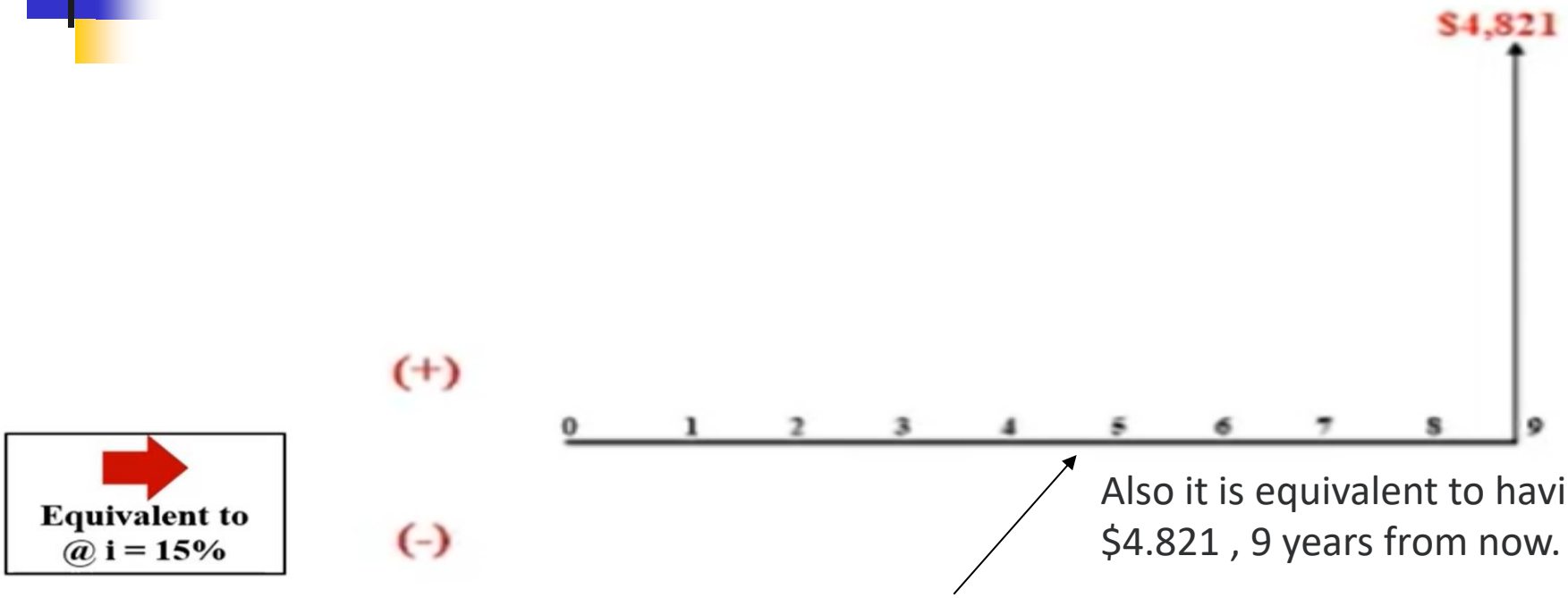
Equivalent to @ i = 15%

(-)

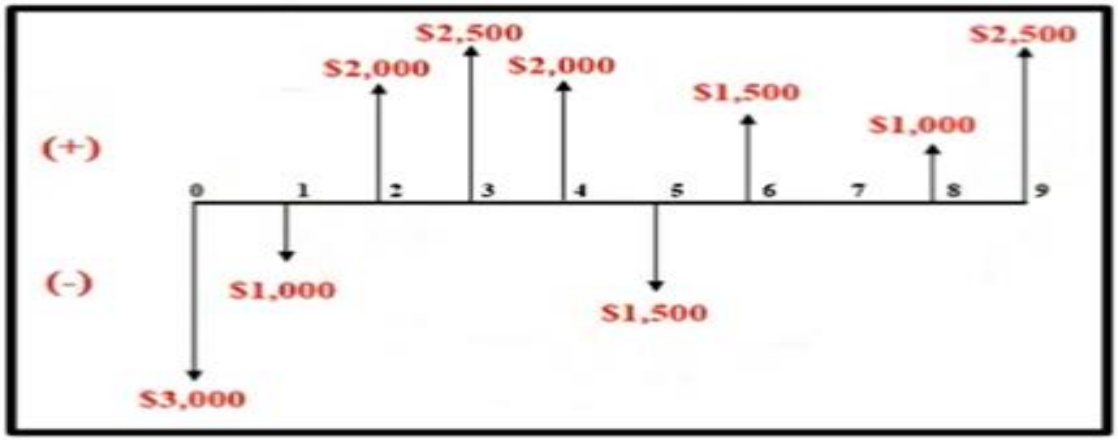
all of these transactions or cash flows are **equivalent** to this cash flow. This applies only if the **interest rate is 15%**



Economic Equivalence

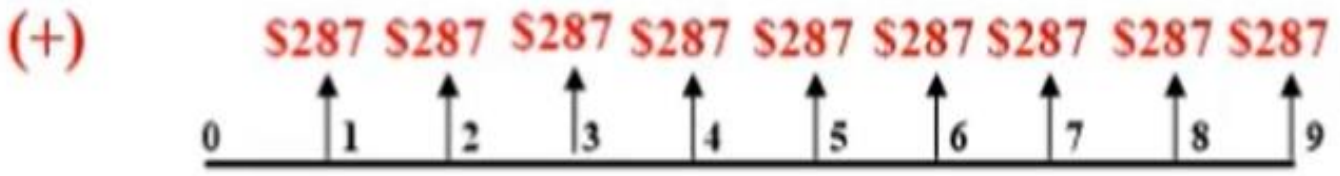


Also it is equivalent to having $4.821 , 9 years from now.



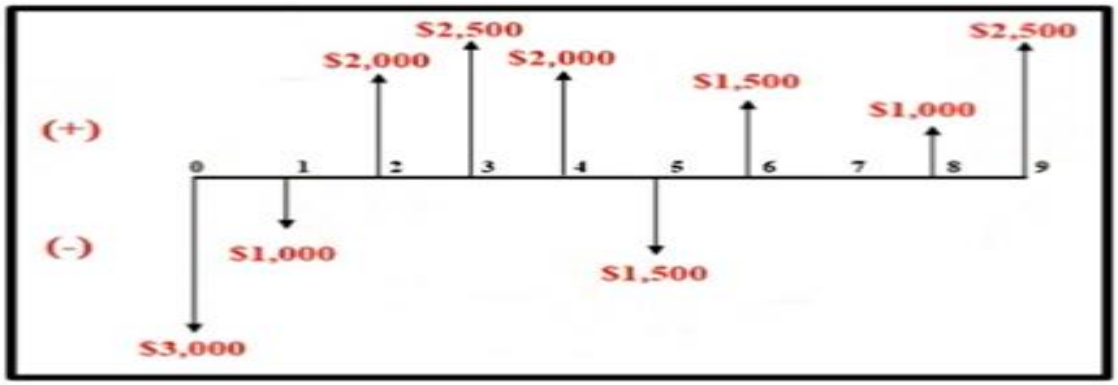
Economic Equivalence





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With the same concept, it's same as if we are having also $287 at the end of each year over the next 9 years

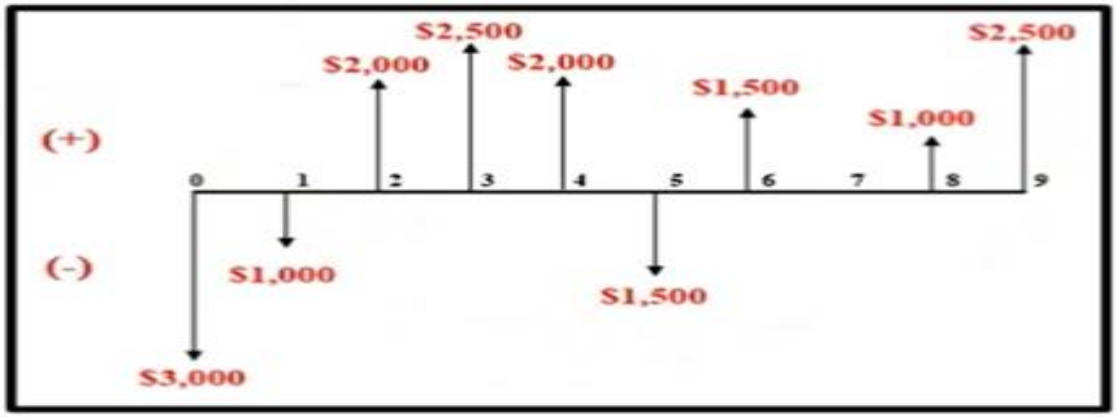


Economic Equivalence

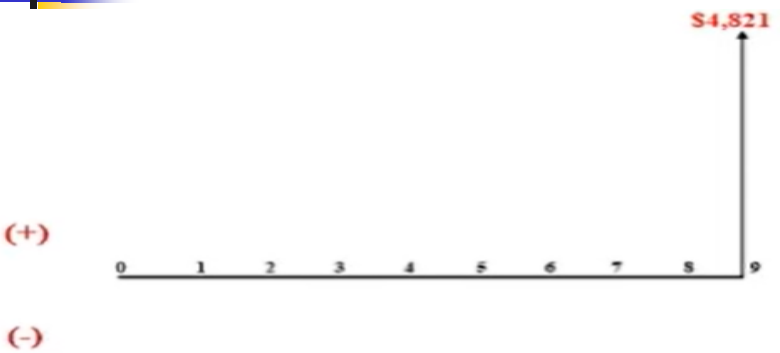


Equivalent to
@ i = 15%

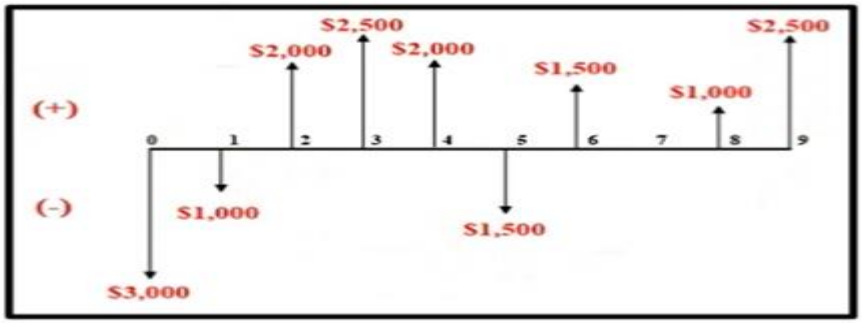
it is equivalent to as if we are having $2.397 dollars at the end of year number 4.

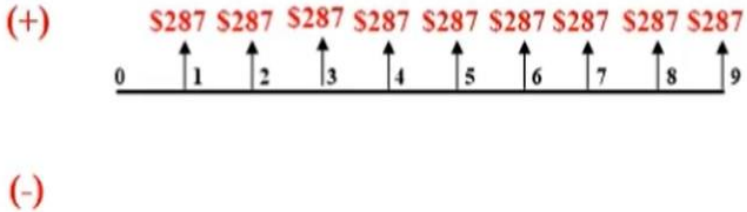


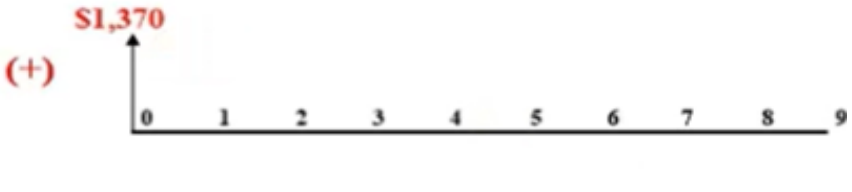
Economic Equivalence











Equivalent to
@ i = 15%

Economic Equivalence

- 1.Single Cash Flow
- 2.Uniform (Equal) Series Cash Flow
- 3.Uneven (Irregular) Series Cash Flow
- 4.Arithmetic (Linear) Gradient Series Cash Flow
- 5.Geometric Gradient Series Cash Flow
- 6.Composite Cash Flow

(1) Single Cash Flow

The simplest case that involves the equivalence of a **single present amount** and its **future worth**

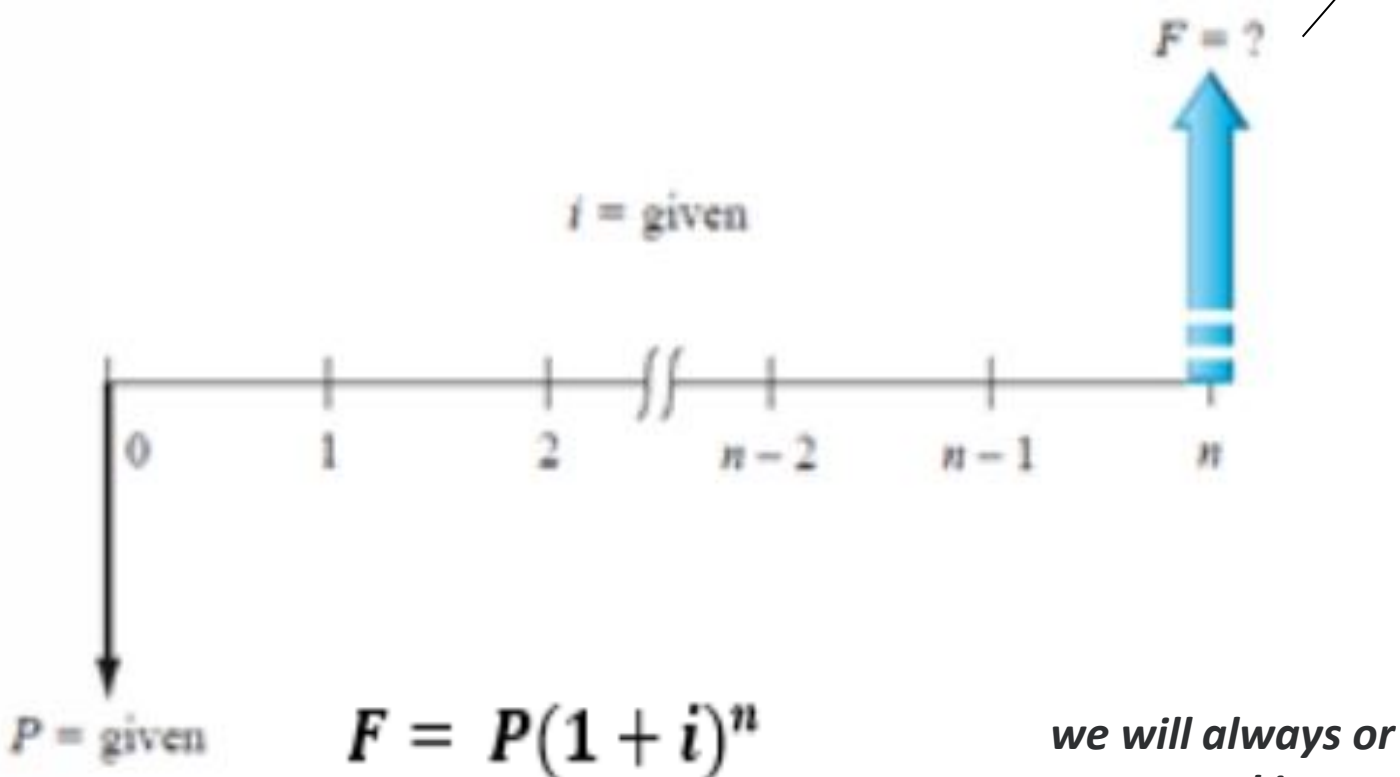
The **equivalence factors** involved in this type of cash flow are:

- **Compounding** Amount Factor (Future Worth)
- **Discounting** Amount Factor (Present Worth)

Single Cash Flow

Compounding Amount Factor:

how much this amount will be in the future amount so we need to **convert** the **present to future.**

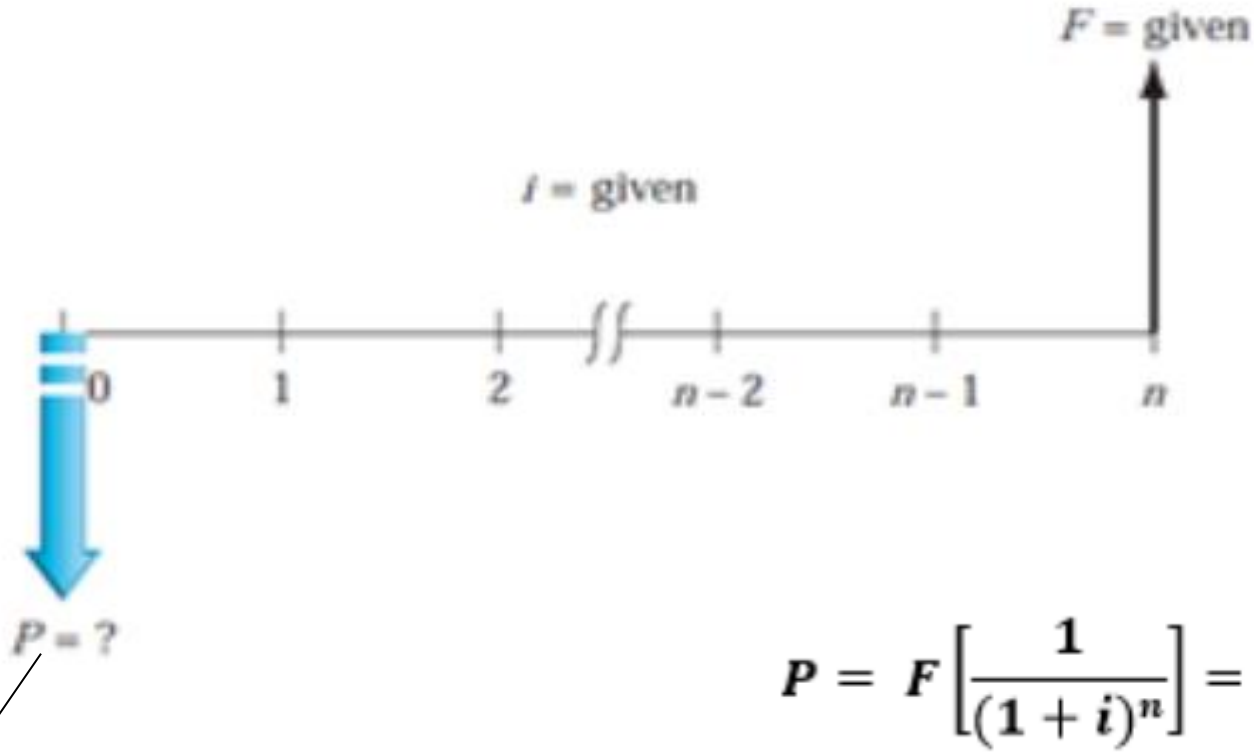


$$F = P(1+i)^n$$

we will always or mostly use the compound interest rate.

Single Cash Flow

Discounting Amount Factor



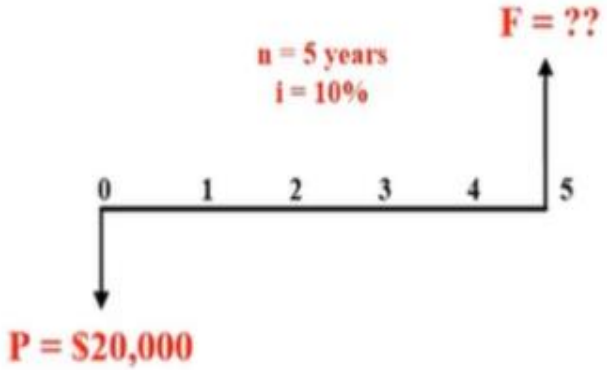
$$P = F\left[\frac{1}{(1+i)^n}\right] = F(1+i)^{-n}$$

how much this amount of **money is worth at the time now**, which is present.

Single Cash Flow

How large will your deposit of $20,000 today be after 5 years considering a 10% interest rate?

P = $20,000, i = 10%, n = 5 years, F = ??

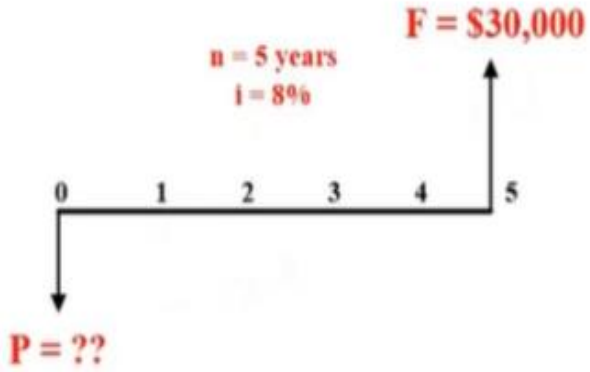


$F = 20000 \times (1 + 0.1)^5 =$ **$32,210.2**

Single Cash Flow

How much should you deposit today if you want to have $30,000 after 5 years considering 8% interest rate?

P = ??, i = 8%, n = 5 years, F = $30,000

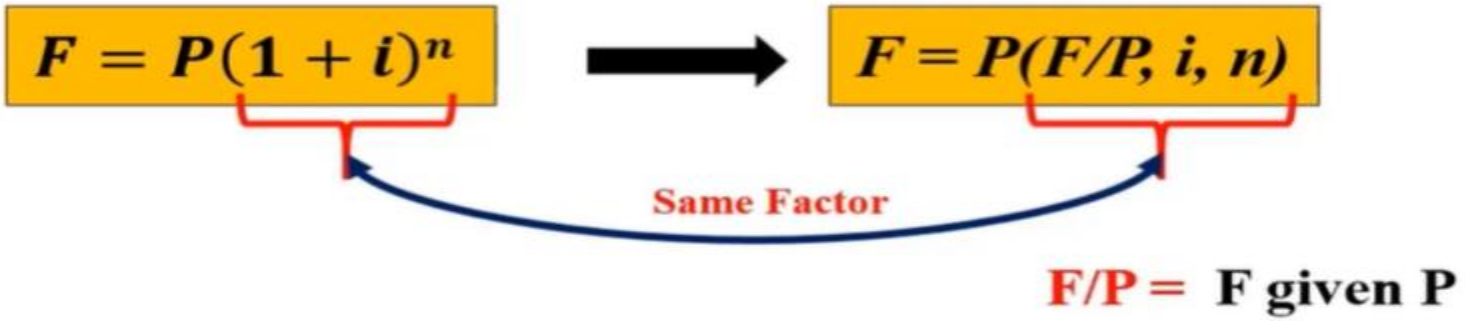


$P = 30000 / (1 + 0.08)^5 = \underline{\$20,417.5}$

You should always know that converting a future to present will result in a value less than the future amount

Single Cash Flow

Compounding Factor:



F/P = F given P

Discounting Factor



P/F = P given F

Single Cash Flow

Solve examples using the interest factors tables

10.0%

	Single Payment		Equal Payment Series				Gradient Series		
N	Compound Amount Factor (F/P,i,N)	Present Worth Factor (P/F,i,N)	Compound Amount Factor (F/A,i,N)	Sinking Fund Factor (A/F,i,N)	Present Worth Factor (P/A,i,N)	Capital Recovery Factor (A/P,i,N)	Gradient Uniform Series (A/G,i,N)	Gradient Present Worth (P/G,i,N)	N
1	1.1000	0.9091	1.0000	1.0000	0.9091	1.1000	0.0000	0.0000	1
2	1.2100	0.8264	2.1000	0.4762	1.7355	0.5762	0.4762	0.8264	2
3	1.3310	0.7513	3.3100	0.3021	2.4869	0.4021	0.9366	2.3291	3
4	1.4641	0.6830	4.6410	0.2155	3.1699	0.3155	1.3812	4.3781	4
5	1.6105	0.6209	6.1051	0.1638	3.7908	0.2638	1.8101	6.8618	5

Example: P = $20.000, i = 10% , n = 5 years, F = ??

$$F = 20000\ (F/P,\ 10\%,\ 5) = 20000 \times 1.6105 = \underline{\$32,210}$$

Single Cash Flow

Solve examples using the interest factors tables

8.0%

	Single Payment		Equal Payment Series				Gradient Series		
N	Compound Amount Factor (F/P,i,N)	Present Worth Factor (P/F,i,N)	Compound Amount Factor (F/A,i,N)	Sinking Fund Factor (A/F,i,N)	Present Worth Factor (P/A,i,N)	Capital Recovery Factor (A/P,i,N)	Gradient Uniform Series (A/G,i,N)	Gradient Present Worth (P/G,i,N)	N
1	1.0800	0.9259	1.0000	1.0000	0.9259	1.0800	0.0000	0.0000	1
2	1.1664	0.8573	2.0800	0.4808	1.7833	0.5608	0.4808	0.8573	2
3	1.2597	0.7938	3.2464	0.3080	2.5771	0.3880	0.9487	2.4450	3
4	1.3605	0.7350	4.5061	0.2219	3.3121	0.3019	1.4040	4.6501	4
5	1.4693	0.6806	5.8666	0.1705	3.9927	0.2505	1.8465	7.3724	5

Example: P = ??, i = 8% , n = 5 years, F = $30.000

$$P = 30000\ (P/F, 8\%, 5) = 30000 \times 0.6806 = \underline{\$20,418}$$

Single Cash Flow & Present Worth

Money travels through time.

1 Single Cash Flow

$$F = P(1 + i)^n$$



Compounding moves money **forward in time.**

> If $20,000 can earn 10%, waiting changes its value.

2 Present Worth

$$P = F(1 + i)^{-n}$$



Discounting translates future money into **today's language.**

> Discounting does not shrink money — it translates it.

Key takeaway: First choose the time point. Then move the money.

(2) Uniform (Equal) Series Cash Flow

- Uniform cash flow transaction involves an equal series of **disbursements (money paid out) or receipts (money received)** (e.g. **installments (regular equal payment) on car loans and home mortgage payments**)

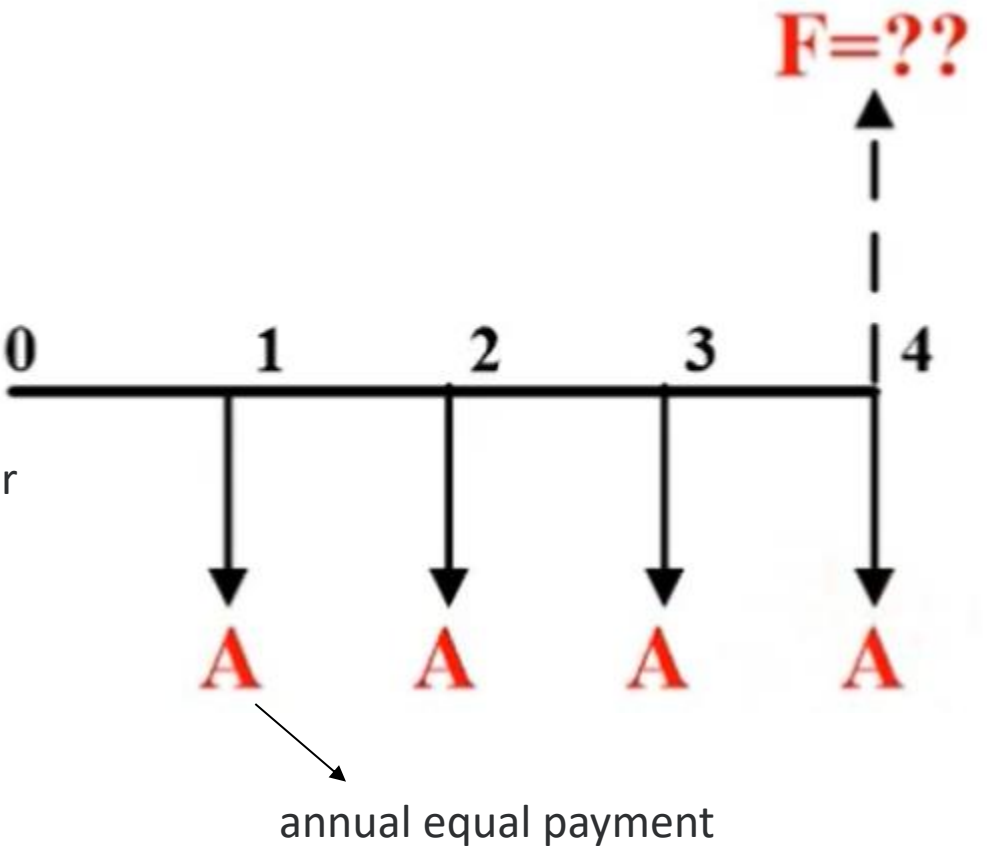
- These cash flow series occur **every period**, period means **every month or every year,** depending on the period you are using in your economic analysis. These **series of cash flows or series of disbursements or receipts** are equal.

- In this type of cash flow, we have 4 main types of equivalence factors.
  - a-) **Uniform** Series **Compounding** Amount Factor (**Future** Worth)
  - b-) **Sinking Fund** Factor
  - c-) **Uniform** Series **Discounting** Amount Factor (**Present** Worth)
  - d-) **Capital Recovery** Factor

Uniform (Equal) Series Cash Flow

a-) Uniform Series Compounding Amount Factor (Future Worth)

(Finding F Given A)

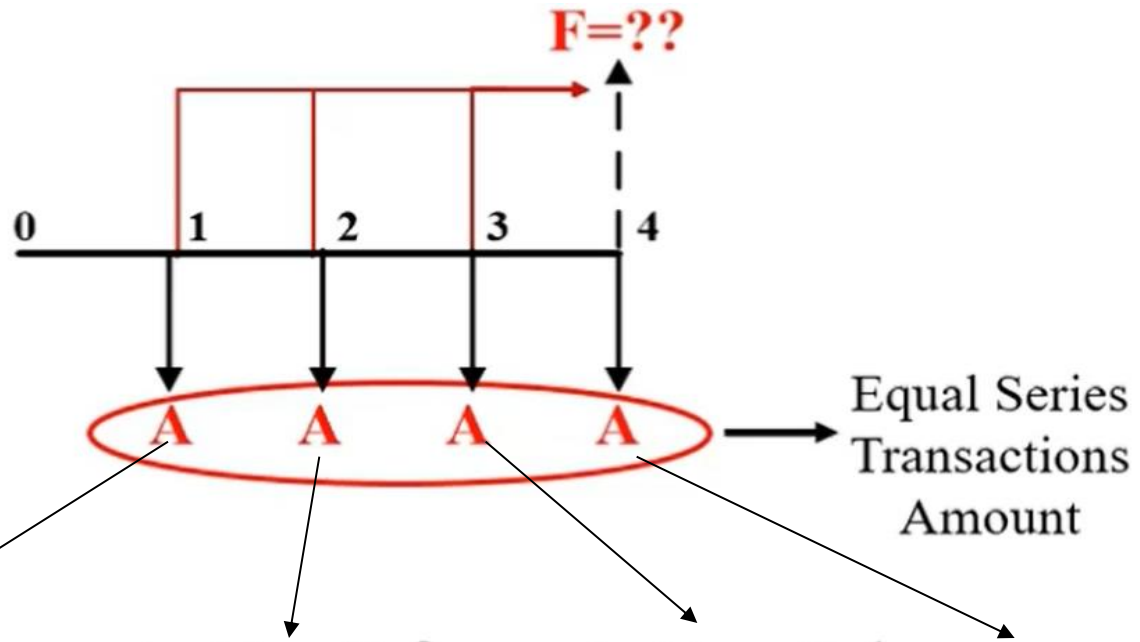


we are having equal series of payments or disbursements in this cash flow
and we need to know how much they're equivalent to as a future amount.
We deal with equals series on a yearly basis.

Uniform (Equal) Series Cash Flow

a-) Uniform Series Compounding Amount Factor (Future Worth)

(Finding F Given A)



$$F = A(1+i)^3 + A(1+i)^2 + A(1+i)^1 + A$$

Uniform (Equal) Series Cash Flow

a-) Uniform Series Compounding Amount Factor (Future Worth)

(Finding **F** Given **A)**

Generally:

$$F = A(1+i)^{n-1} + \ldots + A(1+i)^3 + A(1+i)^2 + A(1+i)^1 + A$$

Multiply both sides by (1+i):

$$F(1+i) = A(1+i)^n + \ldots + A(1+i)^4 + A(1+i)^3 + A(1+i)^2 + A(1+i)^1$$

Factoring out "A" in both equations and subtracting the first equation from the second gives us the following:

$$F(1+i) = A[(1+i)^n + \ldots + (1+i)^4 + (1+i)^3 + (1+i)^2 + (1+i)^1]$$

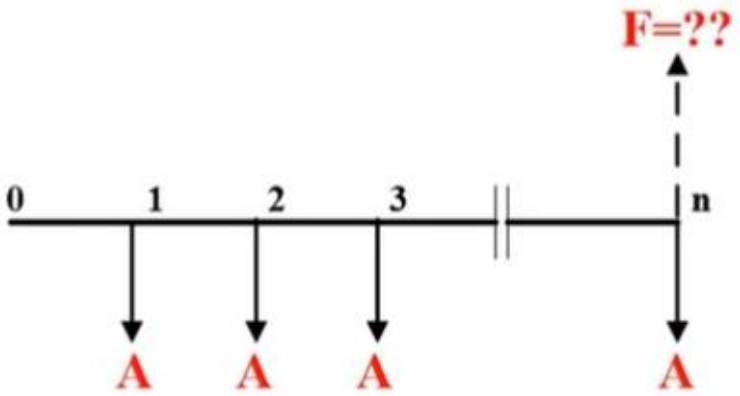
$$- \quad F \quad = A[(1+i)^{n-1} + \ldots + (1+i)^3 + (1+i)^2 + (1+i)^1 + 1]$$

$$Fi = A[(1+i)^n - 1]$$

Uniform (Equal) Series Cash Flow

a-) Uniform Series Compounding Amount Factor (Future Worth)

(Finding F Given A)



*Notice that **F** occurs at the end of the **A** series*

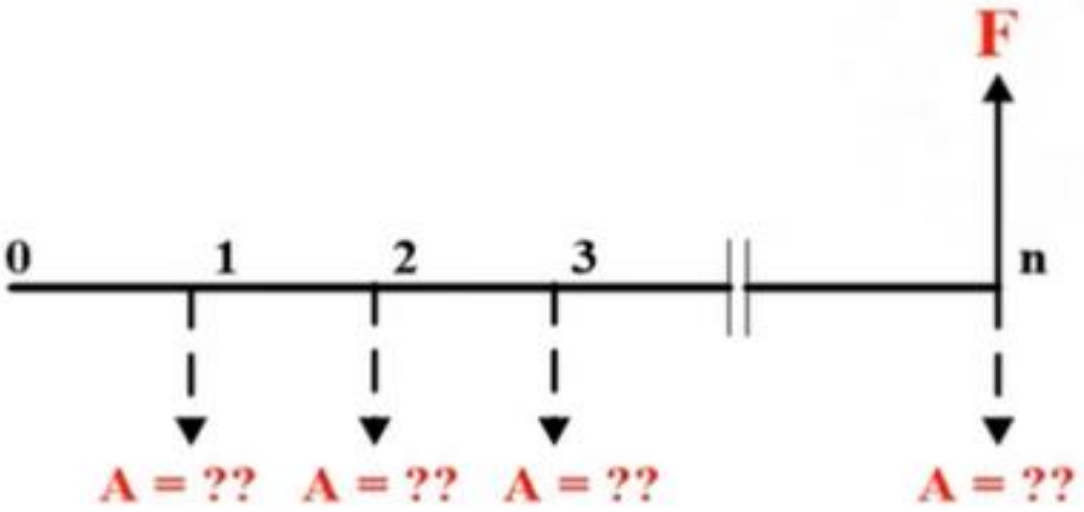
$$F = A\left[\frac{(1+i)^n - 1}{i}\right] \quad \Rightarrow \quad F = A(F/A, i, n)$$

Uniform (Equal) Series Cash Flow

b-) Sinking Fund Factor

(Finding A Given F)

What is equivalent amount of money that we are supposed to deposit in order to achieve such a future amount?



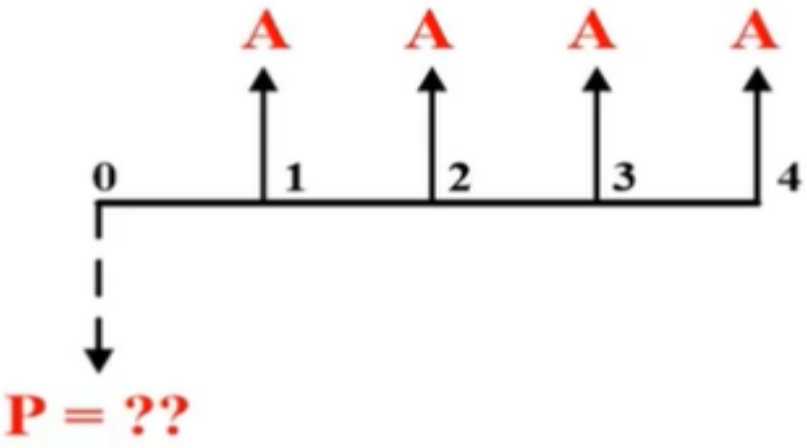
$$A = F\left[\frac{i}{(1+i)^n - 1}\right] \quad \Rightarrow \quad A = F(A/F, i, n)$$

Uniform (Equal) Series Cash Flow

c-) Uniform Series Discounting Amount Factor (Present Worth)

(Finding P Given A)

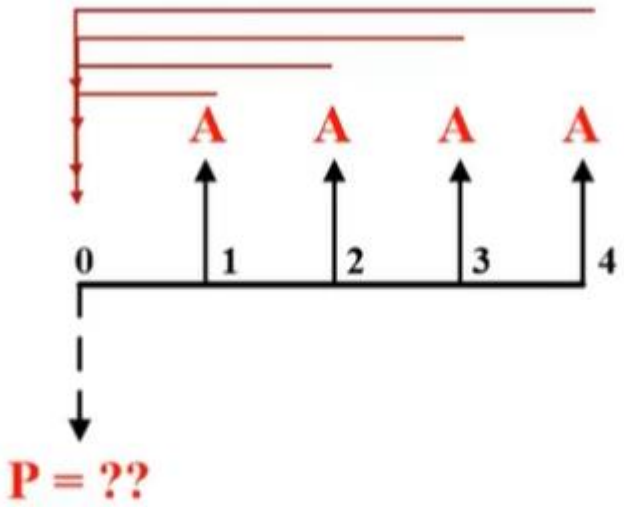
we determine
the present value given
that we have the future
value. So, we're going to
convert all of these A's to
the present time.



Uniform (Equal) Series Cash Flow

c-) Uniform Series Discounting Amount Factor (Present Worth)

(Finding P Given A)



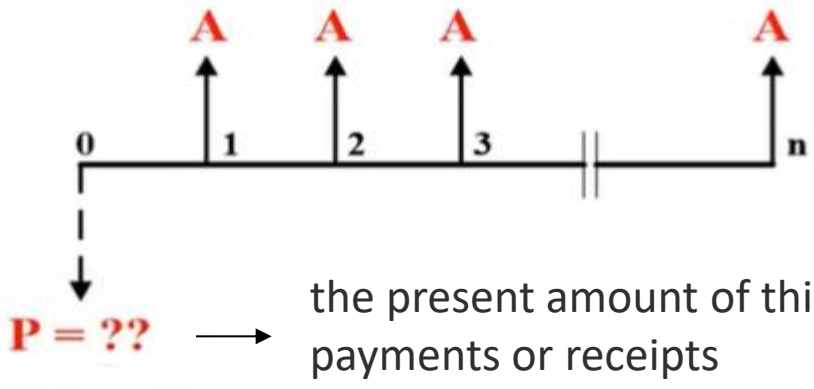
$$P = A(1+i)^{-1} + A(1+i)^{-2} + A(1+i)^{-3} + A(1+i)^{-4}$$

Uniform (Equal) Series Cash Flow

c-) Uniform Series Discounting Amount Factor (Present Worth)

(Finding P Given A)

Notice that **P** occurs one period before the start of the **A** series



the present amount of this series of payments or receipts

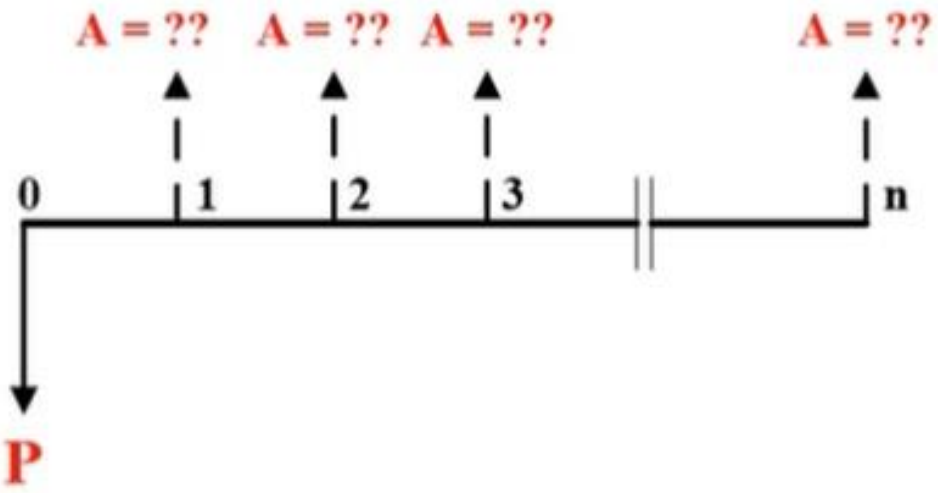
$$P = A\left[\frac{(1+i)^n - 1}{i(1+i)^n}\right] \quad \Rightarrow \quad P = A(P/A, i, n)$$

Uniform (Equal) Series Cash Flow

d-) Capital Recovery Factor

(Finding A Given P)

You have started an initial investment with P and you need to know to recover this investment capital over the next few years.



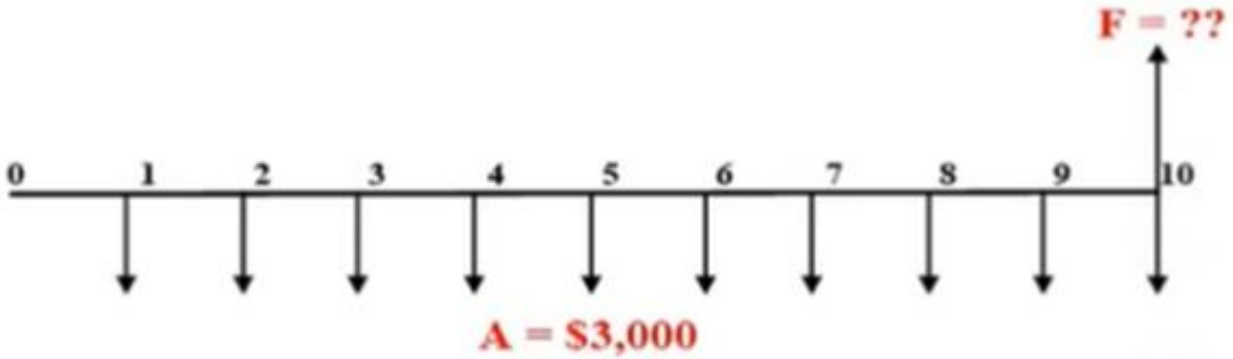
$$A = P\left[\frac{i(1+i)^n}{(1+i)^n - 1}\right] \quad \Rightarrow \quad A = P(A/P, i, n)$$

Uniform (Equal) Series Cash Flow

Example : uniform series compounding amount factor

Suppose you make an annual contribution of $3,000 to your saving account at the end of each year for 10 years. If your saving account earns 7% annual interest rate, how much can be withdrawn at the end of the 10 years?

Usually in engineering economic analysis, if the **time now** is not mentioned we always represent any receipts or any payments or any revenues or expenses that they occur at **the end of the year**.



$$F = 3000(F/A, 7\%, 10) = 3000 \times 13.8164 = \underline{\$41,449.2}$$

Uniform (Equal) Series Cash Flow

Example : uniform series compounding amount factor

7.0%

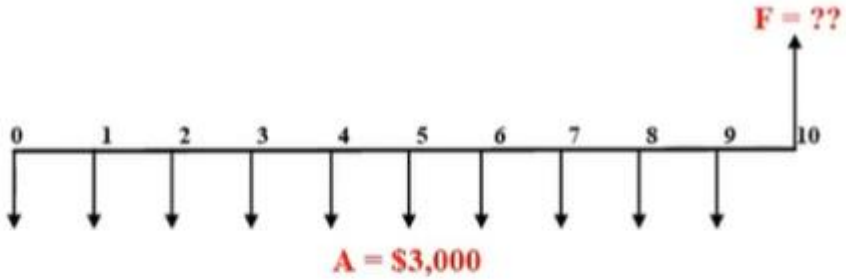
	Single Payment		Equal Payment Series				Gradient Series		
N	Compound Amount Factor *(F/P,i,N)*	Present Worth Factor *(P/F,i,N)*	Compound Amount Factor *(F/A,i,N)*	Sinking Fund Factor *(A/F,i,N)*	Present Worth Factor *(P/A,i,N)*	Capital Recovery Factor *(A/P,i,N)*	Gradient Uniform Series *(A/G,i,N)*	Gradient Present Worth *(P/G,i,N)*	*N*
1	1.0700	0.9346	1.0000	1.0000	0.9346	1.0700	0.0000	0.0000	1
2	1.1449	0.8734	2.0700	0.4831	1.8080	0.5531	0.4831	0.8734	2
3	1.2250	0.8163	3.2149	0.3111	2.6243	0.3811	0.9549	2.5060	3
4	1.3108	0.7629	4.4399	0.2252	3.3872	0.2952	1.4155	4.7947	4
5	1.4026	0.7130	5.7507	0.1739	4.1002	0.2439	1.8650	7.6467	5
6	1.5007	0.6663	7.1533	0.1398	4.7665	0.2098	2.3032	10.9784	6
7	1.6058	0.6227	8.6540	0.1156	5.3893	0.1856	2.7304	14.7149	7
8	1.7182	0.5820	10.2598	0.0975	5.9713	0.1675	3.1465	18.7889	8
9	1.8385	0.5439	11.9780	0.0835	6.5152	0.1535	3.5517	23.1404	9
10	1.9672	0.5083	13.8164	0.0724	7.0236	0.1424	3.9461	27.7156	10

Uniform (Equal) Series Cash Flow

Example : uniform series compounding amount factor

For the previous example, suppose that all the deposits were made at the beginning of each year instead at the end of each year. How much can be withdrawn at the end of the 10 years?

Every deposit was shifted only by one year



$$F_9 = 3000(F/A, 7\%, 10) = 3000 \times 13.8164 = \$41,449.2$$

$$F_{10} = 41449.2(F/P, 7\%, 1) = 41449.2 \times 1.07 = \underline{\$44,350.6}$$

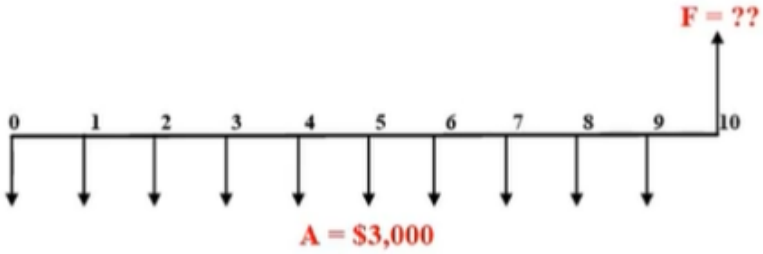
Uniform (Equal) Series Cash Flow

Example : uniform series compounding amount factor

Payment Timing Comparison

Payment No.	End-of-Year Timing	Beginning-of-Year Timing
1st payment	End of Year 1	Beginning of Year 0 (now)
2nd payment	End of Year 2	Beginning of Year 1
...	...	...
9th payment	End of Year 9	Beginning of Year 8
10th payment	End of Year 10	Beginning of Year 9

"Beginning of year" means **each payment occurs one year earlier**, so the **last payment happens at Year 9**, not Year 10.



Uniform (Equal) Series Cash Flow

Example : uniform series compounding amount factor

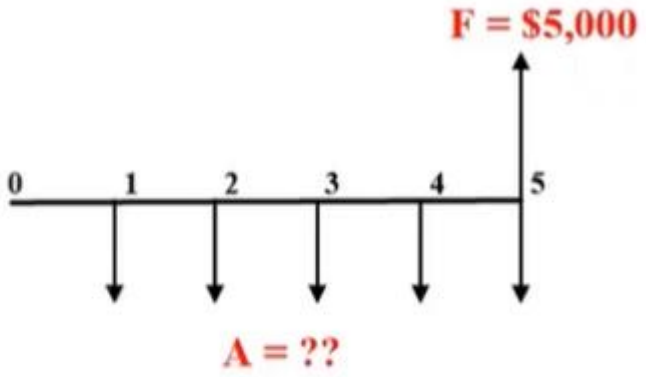
7.0%

	Single Payment		Equal Payment Series				Gradient Series		
N	Compound Amount Factor *(F/P,i,N)*	Present Worth Factor *(P/F,i,N)*	Compound Amount Factor *(F/A,i,N)*	Sinking Fund Factor *(A/F,i,N)*	Present Worth Factor *(P/A,i,N)*	Capital Recovery Factor *(A/P,i,N)*	Gradient Uniform Series *(A/G,i,N)*	Gradient Present Worth *(P/G,i,N)*	*N*
1	1.0700	0.9346	1.0000	1.0000	0.9346	1.0700	0.0000	0.0000	1
2	1.1449	0.8734	2.0700	0.4831	1.8080	0.5531	0.4831	0.8734	2
3	1.2250	0.8163	3.2149	0.3111	2.6243	0.3811	0.9549	2.5060	3
4	1.3108	0.7629	4.4399	0.2252	3.3872	0.2952	1.4155	4.7947	4
5	1.4026	0.7130	5.7507	0.1739	4.1002	0.2439	1.8650	7.6467	5
6	1.5007	0.6663	7.1533	0.1398	4.7665	0.2098	2.3032	10.9784	6
7	1.6058	0.6227	8.6540	0.1156	5.3893	0.1856	2.7304	14.7149	7
8	1.7182	0.5820	10.2598	0.0975	5.9713	0.1675	3.1465	18.7889	8
9	1.8385	0.5439	11.9780	0.0835	6.5152	0.1535	3.5517	23.1404	9
10	1.9672	0.5083	13.8164	0.0724	7.0236	0.1424	3.9461	27.7156	10

Uniform (Equal) Series Cash Flow

Example : sinking fund factor

You are planning to make 5 equal deposits in your account at the end of each year. Your goal is to have $5,000 after 5 years from now. How large must your annual deposit be if your bank pays 7% interest rate?



$$A = 5000(A/F, 7\%, 5) = 5000 \times 0.1739 = \underline{\$869.5}$$

Uniform (Equal) Series Cash Flow

Example : sinking fund factor

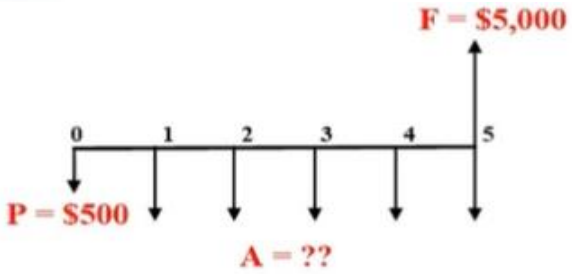
7.0%

	Single Payment		Equal Payment Series				Gradient Series		
N	Compound Amount Factor *(F/P,i,N)*	Present Worth Factor *(P/F,i,N)*	Compound Amount Factor *(F/A,i,N)*	Sinking Fund Factor *(A/F,i,N)*	Present Worth Factor *(P/A,i,N)*	Capital Recovery Factor *(A/P,i,N)*	Gradient Uniform Series *(A/G,i,N)*	Gradient Present Worth *(P/G,i,N)*	*N*
1	1.0700	0.9346	1.0000	1.0000	0.9346	1.0700	0.0000	0.0000	1
2	1.1449	0.8734	2.0700	0.4831	1.8080	0.5531	0.4831	0.8734	2
3	1.2250	0.8163	3.2149	0.3111	2.6243	0.3811	0.9549	2.5060	3
4	1.3108	0.7629	4.4399	0.2252	3.3872	0.2952	1.4155	4.7947	4
5	1.4026	0.7130	5.7507	0.1739	4.1002	0.2439	1.8650	7.6467	5
6	1.5007	0.6663	7.1533	0.1398	4.7665	0.2098	2.3032	10.9784	6
7	1.6058	0.6227	8.6540	0.1156	5.3893	0.1856	2.7304	14.7149	7
8	1.7182	0.5820	10.2598	0.0975	5.9713	0.1675	3.1465	18.7889	8
9	1.8385	0.5439	11.9780	0.0835	6.5152	0.1535	3.5517	23.1404	9
10	1.9672	0.5083	13.8164	0.0724	7.0236	0.1424	3.9461	27.7156	10

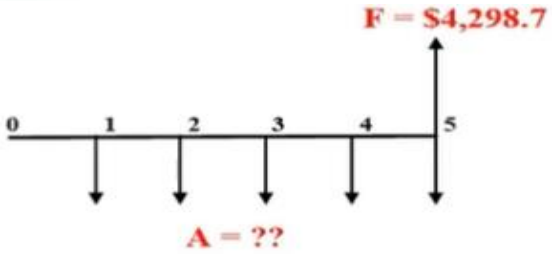
Uniform (Equal) Series Cash Flow

Example : sinking fund factor

For the previous example, suppose that you will deposit now $500 in addition to the 5 equal deposits you are making in your account at the end of each year. Your goal is still the same (i.e. to have $5,000 after 5 years from now). How large must your annual deposit be if your bank pays 7% interest rate?



$$F = 5000 - 500(F/P, 7\%, 5) = 5000 - 701.3 = \underline{\$4,298.7}$$



$$A = 4298.7(A/F, 7\%, 5) = 42987.7 \times 0.1739 = \underline{\$747.5}$$

Uniform (Equal) Series Cash Flow

Example : sinking fund factor

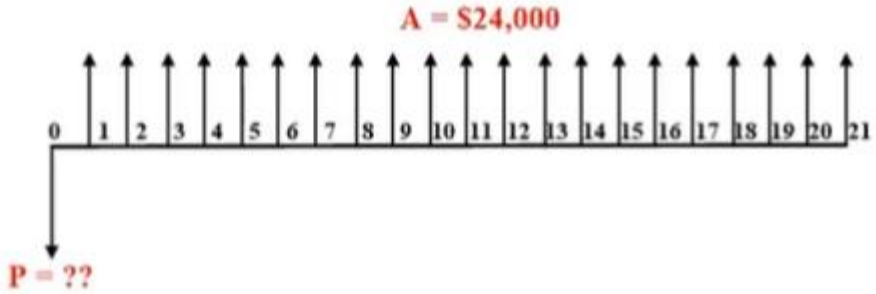
7.0%

	Single Payment		Equal Payment Series				Gradient Series		
N	Compound Amount Factor *(F/P,i,N)*	Present Worth Factor *(P/F,i,N)*	Compound Amount Factor *(F/A,i,N)*	Sinking Fund Factor *(A/F,i,N)*	Present Worth Factor *(P/A,i,N)*	Capital Recovery Factor *(A/P,i,N)*	Gradient Uniform Series *(A/G,i,N)*	Gradient Present Worth *(P/G,i,N)*	*N*
1	1.0700	0.9346	1.0000	1.0000	0.9346	1.0700	0.0000	0.0000	1
2	1.1449	0.8734	2.0700	0.4831	1.8080	0.5531	0.4831	0.8734	2
3	1.2250	0.8163	3.2149	0.3111	2.6243	0.3811	0.9549	2.5060	3
4	1.3108	0.7629	4.4399	0.2252	3.3872	0.2952	1.4155	4.7947	4
5	1.4026	0.7130	5.7507	0.1739	4.1002	0.2439	1.8650	7.6467	5
6	1.5007	0.6663	7.1533	0.1398	4.7665	0.2098	2.3032	10.9784	6
7	1.6058	0.6227	8.6540	0.1156	5.3893	0.1856	2.7304	14.7149	7
8	1.7182	0.5820	10.2598	0.0975	5.9713	0.1675	3.1465	18.7889	8
9	1.8385	0.5439	11.9780	0.0835	6.5152	0.1535	3.5517	23.1404	9
10	1.9672	0.5083	13.8164	0.0724	7.0236	0.1424	3.9461	27.7156	10

Uniform (Equal) Series Cash Flow

Example : Uniform Series Discounting Amount Factor (Present Worth)

You won a lottery and you will receive your winning ticket in 21 annual payments of $24, 000. You decided to start your own business, which required you to secure $250,000 bank loan. You offered to put your future lottery earning to secure the loan. If the bank's interest rate is 10% per year, can you secure the $250,000? If not, how much extra amount of money will you need now to secure the loan?



$P = 24000(P/A, 10\%, 21) = 24000 \times 8.6487 = \$207,569$

$P < \$250,000$ >> **Cannot secure the loan fully**

Extra Amount Required = 250000 – 207569 = **$42,431**

Uniform (Equal) Series Cash Flow

Example : Uniform Series Discounting Amount Factor (Present Worth)

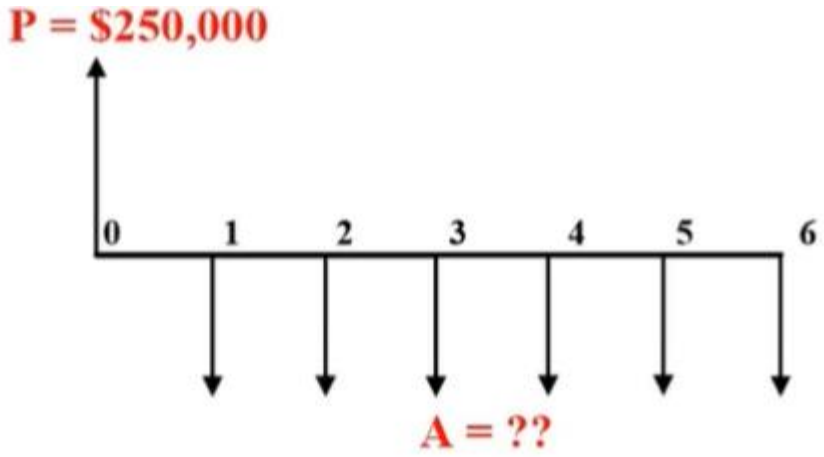
10.0%

	Single Payment		Equal Payment Series				Gradient Series		
N	Compound Amount Factor (F/P,i,N)	Present Worth Factor (P/F,i,N)	Compound Amount Factor (F/A,i,N)	Sinking Fund Factor (A/F,i,N)	Present Worth Factor (P/A,i,N)	Capital Recovery Factor (A/P,i,N)	Gradient Uniform Series (A/G,i,N)	Gradient Present Worth (P/G,i,N)	N
1	1.1000	0.9091	1.0000	1.0000	0.9091	1.1000	0.0000	0.0000	1
2	1.2100	0.8264	2.1000	0.4762	1.7355	0.5762	0.4762	0.8264	2
3	1.3310	0.7513	3.3100	0.3021	2.4869	0.4021	0.9366	2.3291	3
4	1.4641	0.6830	4.6410	0.2155	3.1699	0.3155	1.3812	4.3781	4
5	1.6105	0.6209	6.1051	0.1638	3.7908	0.2638	1.8101	6.8618	5
6	1.7716	0.5645	7.7156	0.1296	4.3553	0.2296	2.2236	9.6842	6
7	1.9487	0.5132	9.4872	0.1054	4.8684	0.2054	2.6216	12.7631	7
8	2.1436	0.4665	11.4359	0.0874	5.3349	0.1874	3.0045	16.0287	8
9	2.3579	0.4241	13.5795	0.0736	5.7590	0.1736	3.3724	19.4215	9
10	2.5937	0.3855	15.9374	0.0627	6.1446	0.1627	3.7255	22.8913	10
11	2.8531	0.3505	18.5312	0.0540	6.4951	0.1540	4.0641	26.3963	11
12	3.1384	0.3186	21.3843	0.0468	6.8137	0.1468	4.3884	29.9012	12
13	3.4523	0.2897	24.5227	0.0408	7.1034	0.1408	4.6988	33.3772	13
14	3.7975	0.2633	27.9750	0.0357	7.3667	0.1357	4.9955	36.8005	14
15	4.1772	0.2394	31.7725	0.0315	7.6061	0.1315	5.2789	40.1520	15
16	4.5950	0.2176	35.9497	0.0278	7.8237	0.1278	5.5493	43.4164	16
17	5.0545	0.1978	40.5447	0.0247	8.0216	0.1247	5.8071	46.5819	17
18	5.5599	0.1799	45.5992	0.0219	8.2014	0.1219	6.0526	49.6395	18
19	6.1159	0.1635	51.1591	0.0195	8.3649	0.1195	6.2861	52.5827	19
20	6.7275	0.1486	57.2750	0.0175	8.5136	0.1175	6.5081	55.4069	20
21	7.4002	0.1351	64.0025	0.0156	8.6487	0.1156	6.7189	58.1095	21
22	8.1403	0.1228	71.4027	0.0140	8.7715	0.1140	6.9189	60.6893	22
23	8.9543	0.1117	79.5430	0.0126	8.8832	0.1126	7.1085	63.1462	23
24	9.8497	0.1015	88.4973	0.0113	8.9847	0.1113	7.2881	65.4813	24
25	10.8347	0.0923	98.3471	0.0102	9.0770	0.1102	7.4580	67.6964	25

Uniform (Equal) Series Cash Flow

Example: Capital Recovery Factor

You borrowed $250,000 to start a business. The loan carries an interest rate of 8% per year and is to be repaid in equal payments over the next 6 years. Compute the amount of this annual payments.



$$A = 250000(A/P, 8\%, 6) = 250000 \times 0.2163 = \underline{\$54,075}$$

Uniform (Equal) Series Cash Flow

Example: Capital Recovery Factor

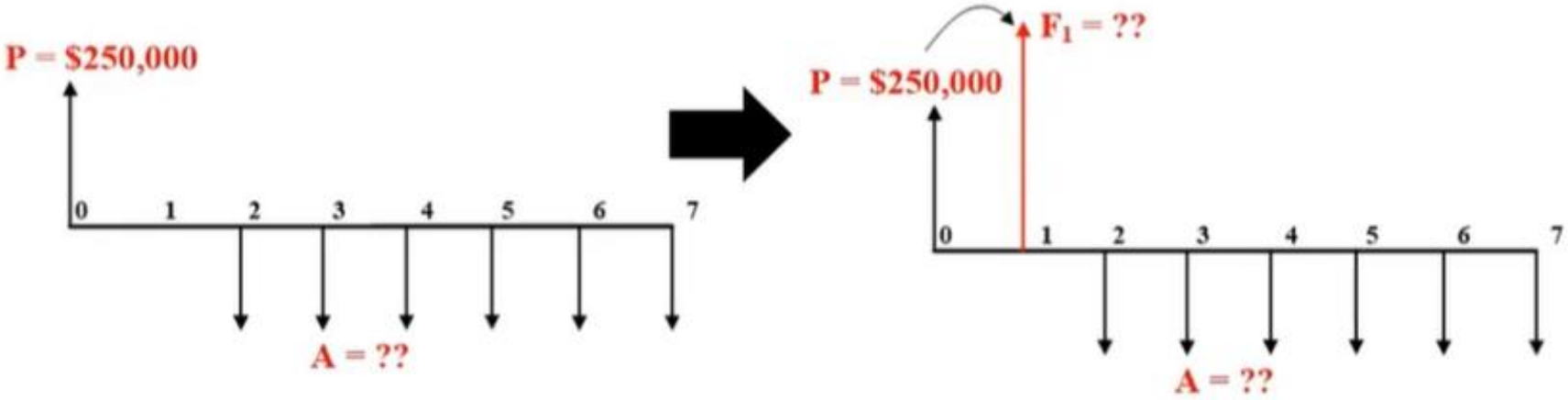
8.0%

	Single Payment		Equal Payment Series				Gradient Series		
N	Compound Amount Factor *(F/P,i,N)*	Present Worth Factor *(P/F,i,N)*	Compound Amount Factor *(F/A,i,N)*	Sinking Fund Factor *(A/F,i,N)*	Present Worth Factor *(P/A,i,N)*	Capital Recovery Factor *(A/P,i,N)*	Gradient Uniform Series *(A/G,i,N)*	Gradient Present Worth *(P/G,i,N)*	*N*
1	1.0800	0.9259	1.0000	1.0000	0.9259	1.0800	0.0000	0.0000	1
2	1.1664	0.8573	2.0800	0.4808	1.7833	0.5608	0.4808	0.8573	2
3	1.2597	0.7938	3.2464	0.3080	2.5771	0.3880	0.9487	2.4450	3
4	1.3605	0.7350	4.5061	0.2219	3.3121	0.3019	1.4040	4.6501	4
5	1.4693	0.6806	5.8666	0.1705	3.9927	0.2505	1.8465	7.3724	5
6	1.5869	0.6302	7.3359	0.1363	4.6229	0.2163	2.2763	10.5233	6
7	1.7138	0.5835	8.9228	0.1121	5.2064	0.1921	2.6937	14.0242	7
8	1.8509	0.5403	10.6366	0.0940	5.7466	0.1740	3.0985	17.8061	8
9	1.9990	0.5002	12.4876	0.0801	6.2469	0.1601	3.4910	21.8081	9
10	2.1589	0.4632	14.4866	0.0690	6.7101	0.1490	3.8713	25.9768	10

Uniform (Equal) Series Cash Flow

Example: Capital Recovery Factor

For the previous example, suppose that the bank agreed to defer the first loan repayment until the end of year 2. Compute the new amount of the annual payments.



$F_1 = 250000(F/P, 8\%, 1) = 250000 \times 1.08 = \$270,000$

$A = 270000(A/P, 8\%, 6) = 270000 \times 0.2163 = \underline{\$58,401}$

Uniform (Equal) Series Cash Flow

Example: Capital Recovery Factor

8.0%

	Single Payment		Equal Payment Series				Gradient Series		
N	Compound Amount Factor *(F/P,i,N)*	Present Worth Factor *(P/F,i,N)*	Compound Amount Factor *(F/A,i,N)*	Sinking Fund Factor *(A/F,i,N)*	Present Worth Factor *(P/A,i,N)*	Capital Recovery Factor *(A/P,i,N)*	Gradient Uniform Series *(A/G,i,N)*	Gradient Present Worth *(P/G,i,N)*	*N*
1	1.0800	0.9259	1.0000	1.0000	0.9259	1.0800	0.0000	0.0000	1
2	1.1664	0.8573	2.0800	0.4808	1.7833	0.5608	0.4808	0.8573	2
3	1.2597	0.7938	3.2464	0.3080	2.5771	0.3880	0.9487	2.4450	3
4	1.3605	0.7350	4.5061	0.2219	3.3121	0.3019	1.4040	4.6501	4
5	1.4693	0.6806	5.8666	0.1705	3.9927	0.2505	1.8465	7.3724	5
6	1.5869	0.6302	7.3359	0.1363	4.6229	0.2163	2.2763	10.5233	6
7	1.7138	0.5835	8.9228	0.1121	5.2064	0.1921	2.6937	14.0242	7
8	1.8509	0.5403	10.6366	0.0940	5.7466	0.1740	3.0985	17.8061	8
9	1.9990	0.5002	12.4876	0.0801	6.2469	0.1601	3.4910	21.8081	9
10	2.1589	0.4632	14.4866	0.0690	6.7101	0.1490	3.8713	25.9768	10

Which is Better?



$1,000
today

or



$1,200
in one year

At first glance, $1,200 looks better.
But in engineering economics, we look at **time** as well.

A Startup Decision



Buy now
$50,000

or



Pay every year
$10,000 per year

Which option is cheaper?
You cannot just compare total amounts.

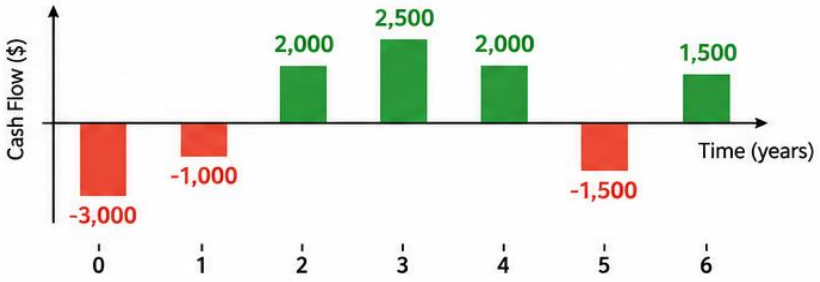
$10,000 in 2026 is not the same as $10,000 in 2031.
Money at different times is not the same economic object.

Uniform Series in Real Life



A uniform series is a set of **equal payments** at **regular time intervals**.

Uneven Cash Flow in a Startup



Real life cash flows are often **uneven.**
We can still convert all these cash flows to a **single equivalent value** at a common point in time.

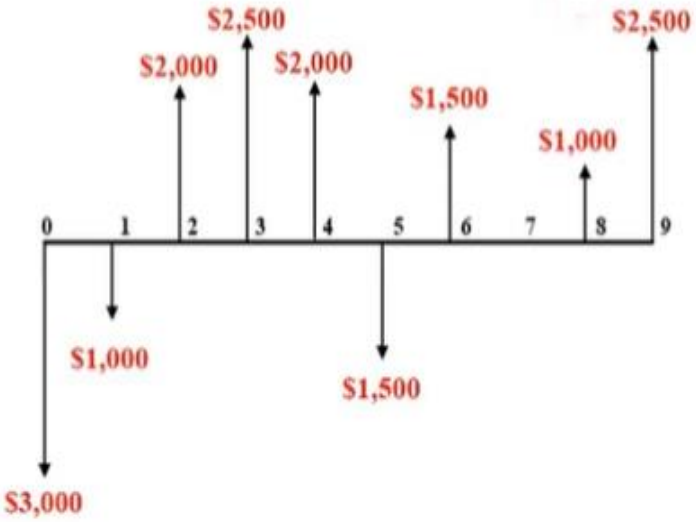
(3) Uneven (Irregular) Series Cash Flow

- A common cash flow transaction involves a series of disbursements or receipts
- For example, series payments are payment of installments on car loans and home mortgage payments. These series payments on car loans or home mortgages typically involve sums to be paid at regular intervals
- However, when there is no clear pattern over the series, we call the transaction an ***uneven cash flow series***

Uneven (Irregular) Series Cash Flow

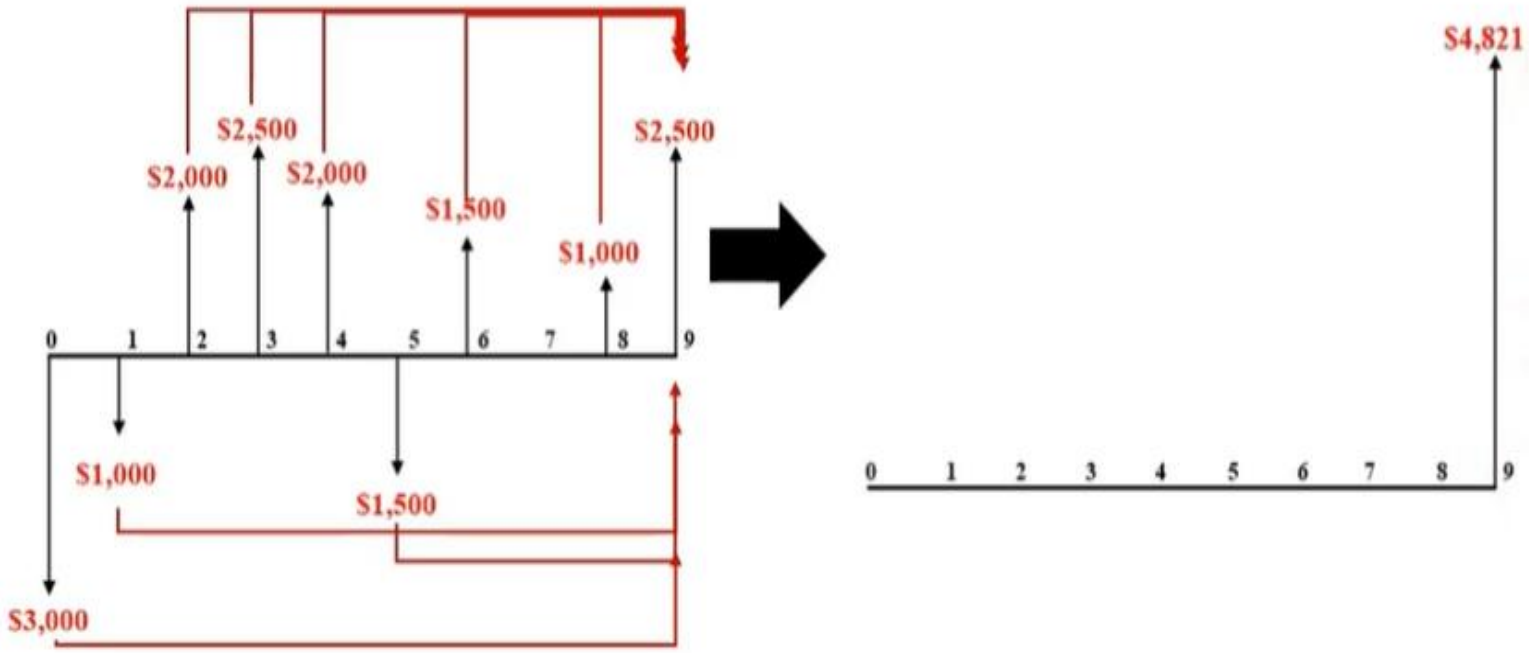
For the irregular series of transactions shown in the figure below and considering a 15% interest rate, determine the following:

(a) The "F" value
(b) The "P" value
(c) The "A" value
(d) The Equivalent Value "EV" at the end of year 4 (i.e. @t = 4)



Uneven (Irregular) Series Cash Flow

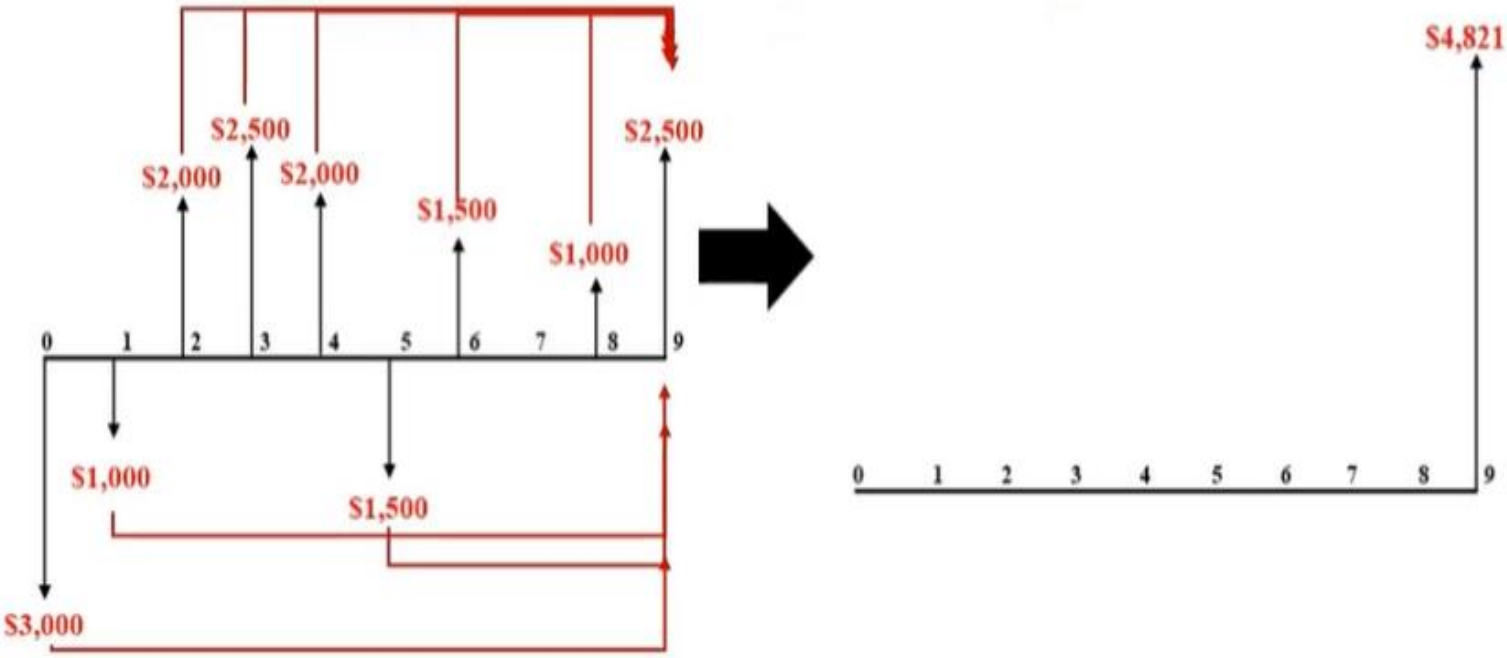
$$\textbf{(a) } F = 2500 + 1000(1.15)^1 + 1500(1.15)^3 - 1500(1.15)^4 + 2000(1.15)^5 + 2500(1.15)^6 + 2000(1.15)^7 - 1000(1.15)^8 - 3000(1.15)^9 = \underline{+\$4{,}820.6}$$



Uneven (Irregular) Series Cash Flow

(a) $F = 2500 + 1000(F/P, 15\%, 1) + 1500(F/P, 15\%, 3) - 1500(F/P, 15\%, 4) + 2000(F/P, 15\%, 5)$

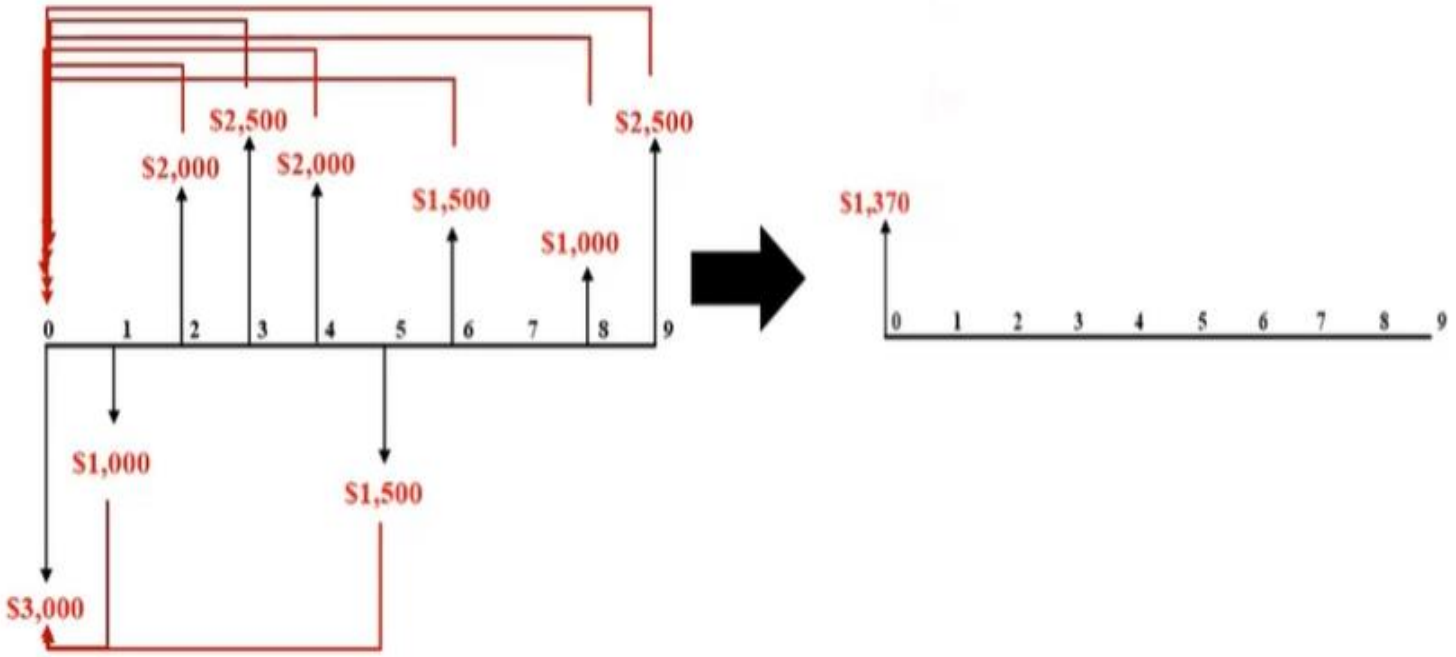
$+ 2500(F/P, 15\%, 6) + 2000(F/P, 15\%, 7) - 1000(F/P, 15\%, 8) - 3000(F/P, 15\%, 9) = \underline{+\$4,820.6}$



Uneven (Irregular) Series Cash Flow

(b) $P = -3000 - 1000(P/F, 15\%, 1) + 2000(P/F, 15\%, 2) + 2500(P/F, 15\%, 3) + 2000(P/F, 15\%, 4)$

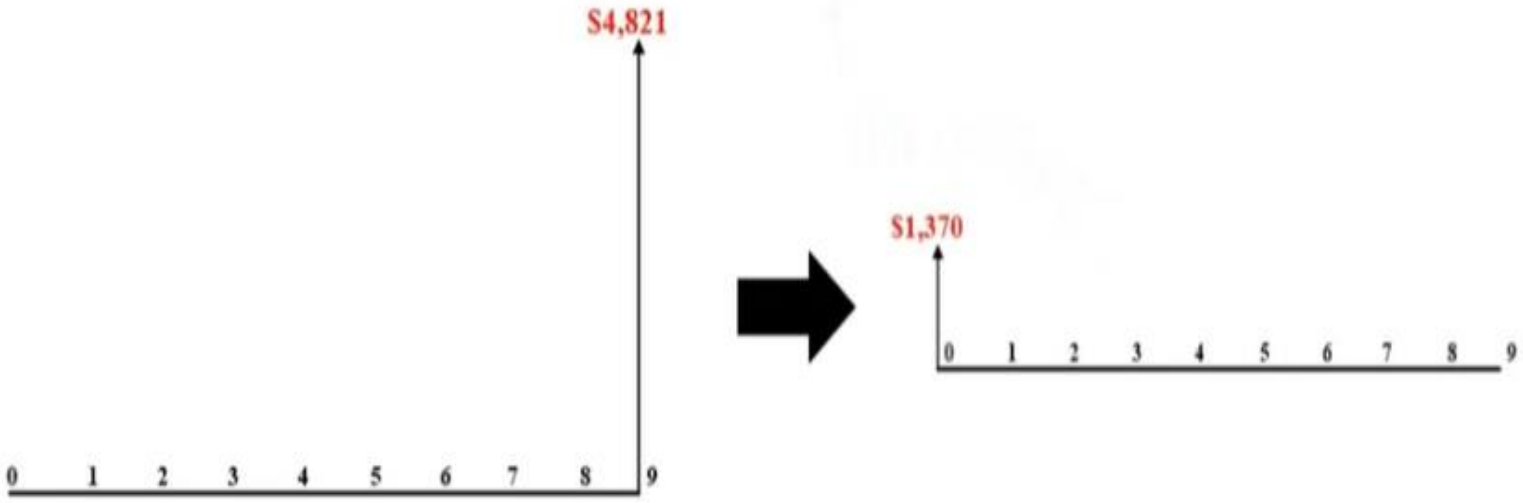
$- 1500(P/F, 15\%, 5) + 1500(P/F, 15\%, 6) + 1000(P/F, 15\%, 8) + 2500(P/F, 15\%, 9) = \underline{+\$1,370.3}$



Uneven (Irregular) Series Cash Flow

(b) We know from (a) that F = $4,820.6

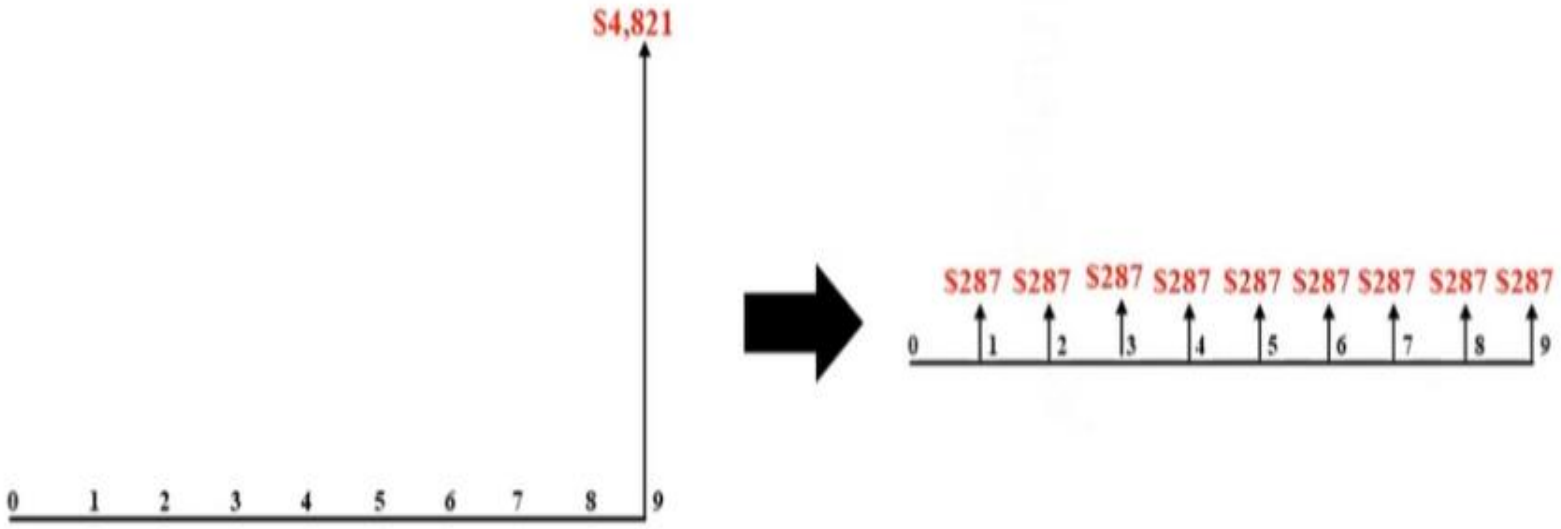
Therefore, $P = F(1+i)^{-n} = 4820.6(1.15)^{-9} =$ **+$1,370.3**



Uneven (Irregular) Series Cash Flow

(c) We know from (a) that F = $4,820.6

Therefore, A = F(A/F, 15%, 9) = 4820.6 x 0.0596 = **+$287.3**



Uneven (Irregular) Series Cash Flow

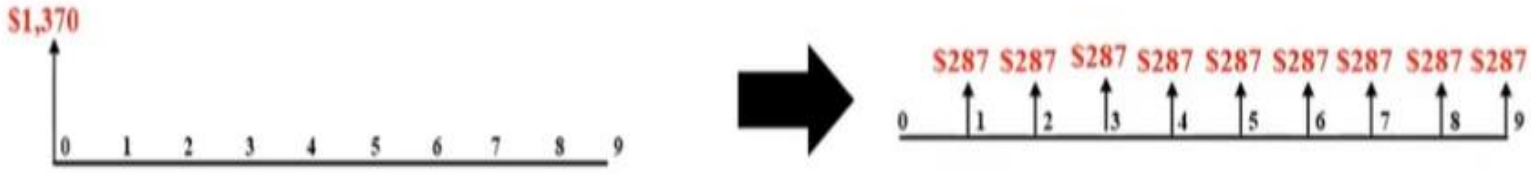
15.0%

	Single Payment		Equal Payment Series				Gradient Series		
N	Compound Amount Factor *(F/P,i,N)*	Present Worth Factor *(P/F,i,N)*	Compound Amount Factor *(F/A,i,N)*	Sinking Fund Factor *(A/F,i,N)*	Present Worth Factor *(P/A,i,N)*	Capital Recovery Factor *(A/P,i,N)*	Gradient Uniform Series *(A/G,i,N)*	Gradient Present Worth *(P/G,i,N)*	*N*
1	1.1500	0.8696	1.0000	1.0000	0.8696	1.1500	0.0000	0.0000	1
2	1.3225	0.7561	2.1500	0.4651	1.6257	0.6151	0.4651	0.7561	2
3	1.5209	0.6575	3.4725	0.2880	2.2832	0.4380	0.9071	2.0712	3
4	1.7490	0.5718	4.9934	0.2003	2.8550	0.3503	1.3263	3.7864	4
5	2.0114	0.4972	6.7424	0.1483	3.3522	0.2983	1.7228	5.7751	5
6	2.3131	0.4323	8.7537	0.1142	3.7845	0.2642	2.0972	7.9368	6
7	2.6600	0.3759	11.0668	0.0904	4.1604	0.2404	2.4498	10.1924	7
8	3.0590	0.3269	13.7268	0.0729	4.4873	0.2229	2.7813	12.4807	8
9	3.5179	0.2843	16.7858	0.0596	4.7716	0.2096	3.0922	14.7548	9
10	4.0456	0.2472	20.3037	0.0493	5.0188	0.1993	3.3832	16.9795	10
11	4.6524	0.2149	24.3493	0.0411	5.2337	0.1911	3.6549	19.1289	11
12	5.3503	0.1869	29.0017	0.0345	5.4206	0.1845	3.9082	21.1849	12
13	6.1528	0.1625	34.3519	0.0291	5.5831	0.1791	4.1438	23.1352	13
14	7.0757	0.1413	40.5047	0.0247	5.7245	0.1747	4.3624	24.9725	14
15	8.1371	0.1229	47.5804	0.0210	5.8474	0.1710	4.5650	26.6930	15

Uneven (Irregular) Series Cash Flow

(c) We know from (b) that P = $1,370.3

Therefore, A = P(A/P, 15%, 9) = 1370.3 x 0.2096 = **+$287.2**



Uneven (Irregular) Series Cash Flow

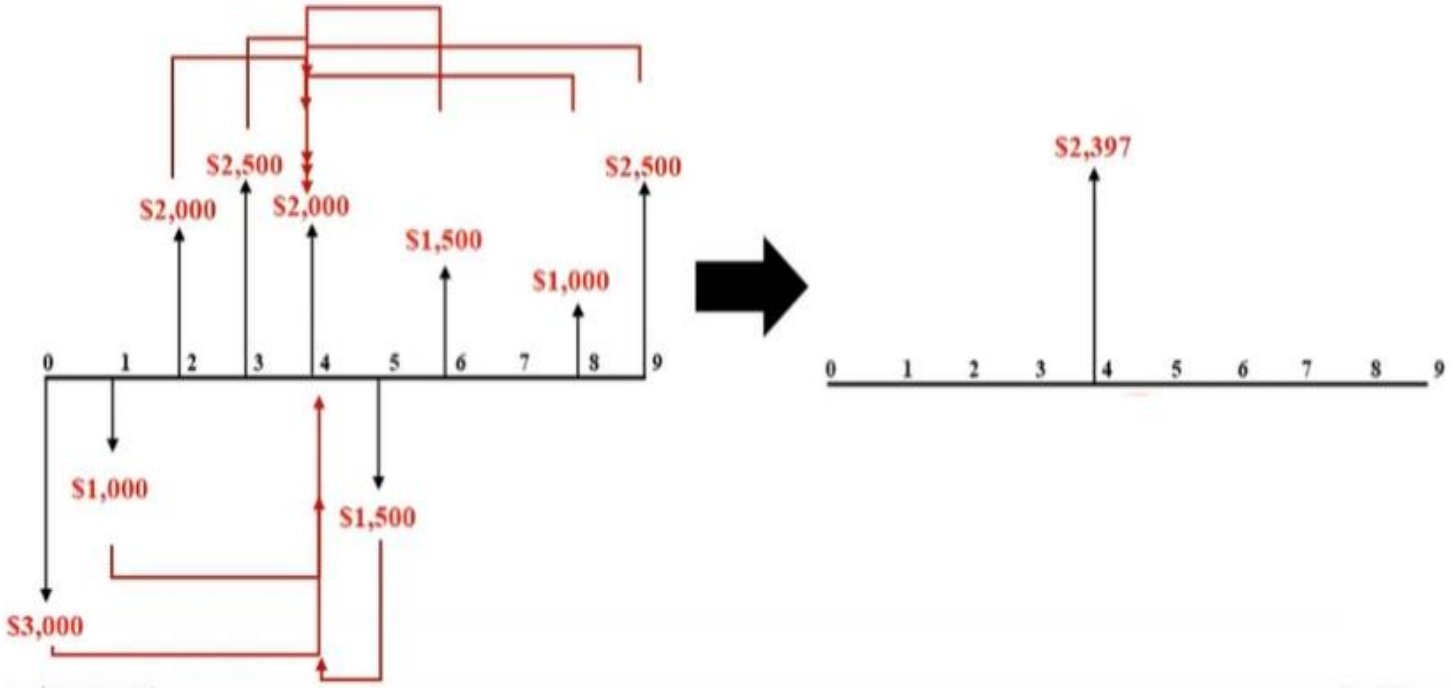
15.0%

	Single Payment		Equal Payment Series				Gradient Series		
N	Compound Amount Factor *(F/P,i,N)*	Present Worth Factor *(P/F,i,N)*	Compound Amount Factor *(F/A,i,N)*	Sinking Fund Factor *(A/F,i,N)*	Present Worth Factor *(P/A,i,N)*	Capital Recovery Factor *(A/P,i,N)*	Gradient Uniform Series *(A/G,i,N)*	Gradient Present Worth *(P/G,i,N)*	*N*
1	1.1500	0.8696	1.0000	1.0000	0.8696	1.1500	0.0000	0.0000	1
2	1.3225	0.7561	2.1500	0.4651	1.6257	0.6151	0.4651	0.7561	2
3	1.5209	0.6575	3.4725	0.2880	2.2832	0.4380	0.9071	2.0712	3
4	1.7490	0.5718	4.9934	0.2003	2.8550	0.3503	1.3263	3.7864	4
5	2.0114	0.4972	6.7424	0.1483	3.3522	0.2983	1.7228	5.7751	5
6	2.3131	0.4323	8.7537	0.1142	3.7845	0.2642	2.0972	7.9368	6
7	2.6600	0.3759	11.0668	0.0904	4.1604	0.2404	2.4498	10.1924	7
8	3.0590	0.3269	13.7268	0.0729	4.4873	0.2229	2.7813	12.4807	8
9	3.5179	0.2843	16.7858	0.0596	4.7716	0.2096	3.0922	14.7548	9
10	4.0456	0.2472	20.3037	0.0493	5.0188	0.1993	3.3832	16.9795	10
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12	5.3503	0.1869	29.0017	0.0345	5.4206	0.1845	3.9082	21.1849	12
13	6.1528	0.1625	34.3519	0.0291	5.5831	0.1791	4.1438	23.1352	13
14	7.0757	0.1413	40.5047	0.0247	5.7245	0.1747	4.3624	24.9725	14
15	8.1371	0.1229	47.5804	0.0210	5.8474	0.1710	4.5650	26.6930	15

Uneven (Irregular) Series Cash Flow

$$\textbf{(d) } EV_4 = 2000 - 3000(1.15)^4 - 1000(1.15)^3 + 2000(1.15)^2 + 2500(1.15)^1 - 1500(1.15)^{-1}$$

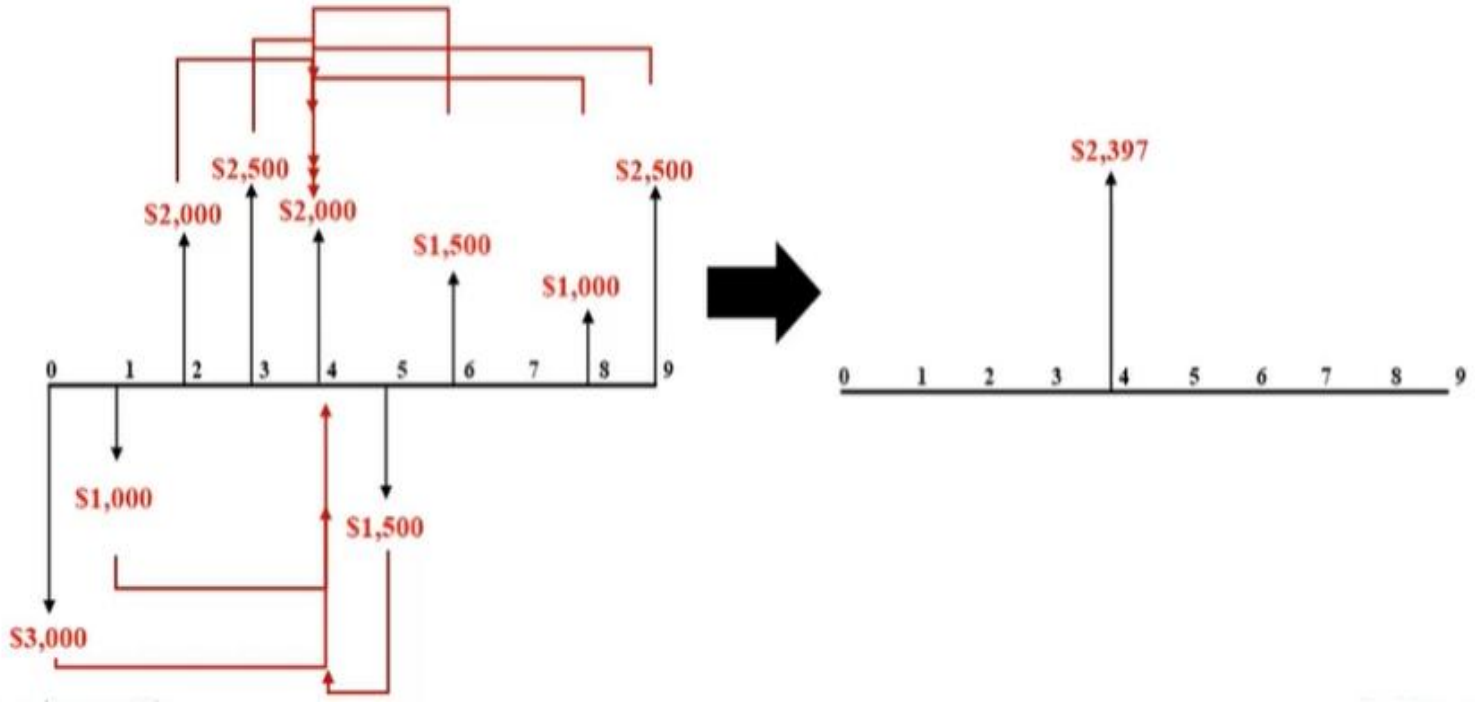
$$+ 1500(1.15)^{-2} + 1000(1.15)^{-4} + 2500(1.15)^{-5} = \underline{+\$2{,}396.7}$$



Uneven (Irregular) Series Cash Flow

(d) $EV_4 = 2000 - 3000(F/P, 15\%, 4) - 1000(F/P, 15\%, 3) + 2000(F/P, 15\%, 2) + 2500(F/P, 15\%, 1)$

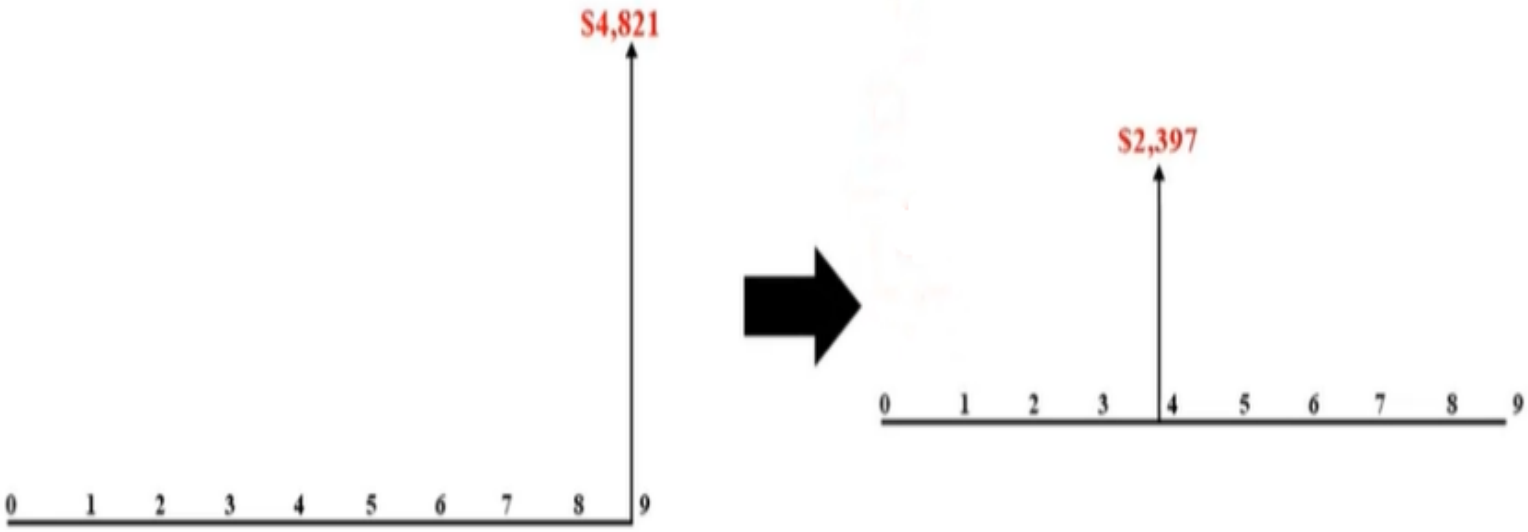
$- 1500(P/F, 15\%, 1) + 1500(P/F, 15\%, 2) + 1000(P/F, 15\%, 4) + 2500(P/F, 15\%, 5) = +\$2,396.7$



Uneven (Irregular) Series Cash Flow

(d) We know from (a) that F = $4,820.6

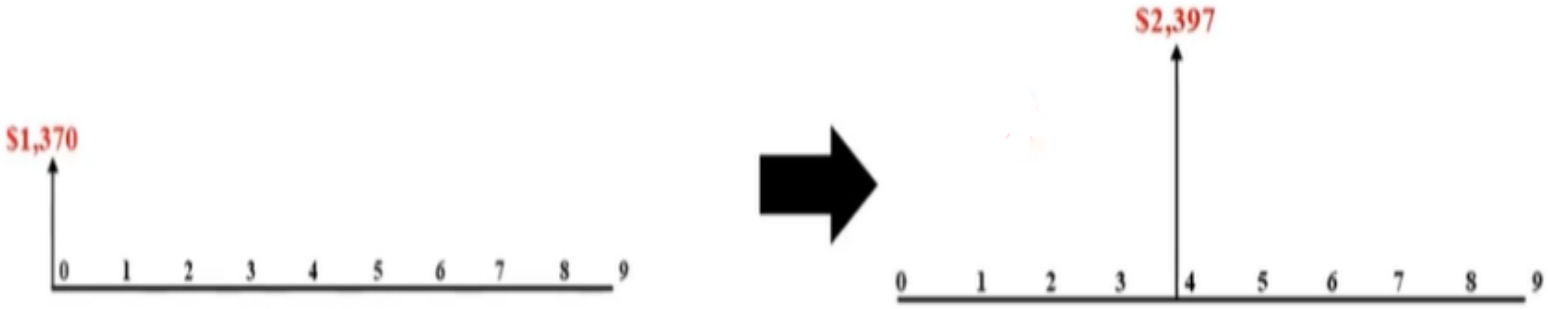
Therefore, $EV_4 = F(1+i)^{-n} = 4820.6(1.15)^{-5} =$ **+$2,396.7**



Uneven (Irregular) Series Cash Flow

(d) We know from (b) that P = $1,370.3

Therefore, $EV_4 = P(1+i)^n = 1370.3(1.15)^4 =$ **+$2,396.7**



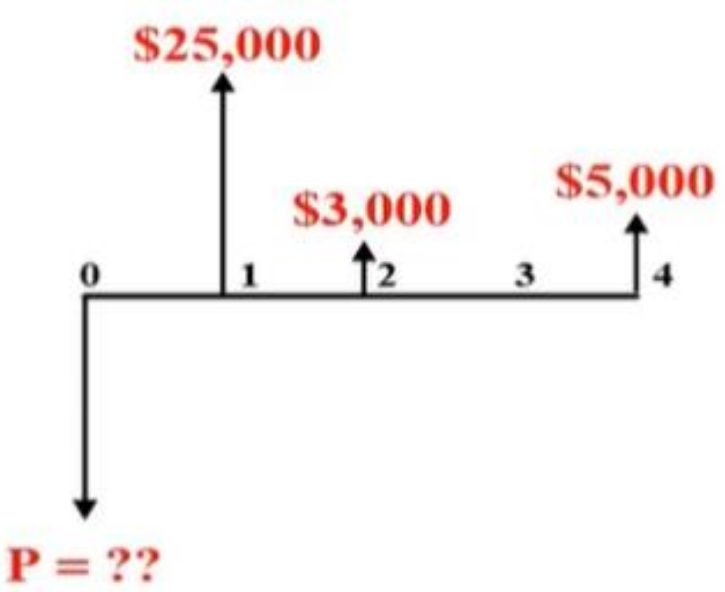
Uneven (Irregular) Series Cash Flow

Wilson Technology, a growing machine shop, wishes to set aside money now to invest over the next 4 years in automating its customer service department. The company can earn 10% on a lump-sum deposited now, and wishes to withdraw the money in the following increments:

Year 1: $25,000 to purchase a computer and database software
Year 2: $3,000 to purchase additional hardware
Year 3: No expenses
Year 4: $5,000 to purchase software upgrades

How much money must be deposited now to cover the anticipated payments over the next 4 years?

Uneven (Irregular) Series Cash Flow



$$P = 25000(P/F, 10\%, 1) + 3000(P/F, 10\%, 2) + 5000(P/F, 10\%, 4)$$

$$= \underline{\$28,621.7}$$

Uneven Cash Flow & The Big Picture

Real life is rarely perfectly uniform

1 Uneven Cash Flow

Real companies do not earn and spend the same amount every year.



One year brings customers. Another year brings a server failure. Another year brings investment.

Can we turn financial noise into one meaningful number?

Yes — by moving every cash flow to a common point in time.

2 Key Rules to Repeat

Never compare money at different times directly.

First choose the time point, then move the money.

Cash flow direction matters.

Economic equivalence is translation across time.

> Engineering economics is not really about formulas. It is about making fair comparisons between choices that happen at different moments in time.